

GRID-CLOCK QUANTUM GRAVITY

A Finite Strict-Local Quantum Cellular Automaton Framework for Emergent Spacetime, Einstein Gravity, Regular Black Holes, Rotating Strong Gravity, and Finite Unitary Radiation

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STATUS NOTE. This document is designed to stand alone. It integrates the finite-QCA construction, smooth-limit gravity program, spin-2 sector, strong-gravity audits, rotating branch, finite radiation carrier, evaporation analysis, and the final E56 claim audit in one volume. The verification program E1-E56 is complete as a research program. The result is an advanced candidate framework, not a completed or unique theory of quantum gravity. Exact finite constructions, infrared derivations, matched parameters, effective/conditional models, and open problems are labeled separately throughout.

Abstract

Grid-Clock Quantum Gravity (GCC) is developed as a finite-information candidate framework in which a fixed body-centered-cubic (BCC) relational graph and a universal microscopic update index are more fundamental than continuum spacetime. The purpose of this integrated paper is to test how far one strict-local finite quantum cellular automaton (QCA) architecture can be pushed toward relativistic matter, Einstein gravity, regular strong-gravity behavior, rotating black holes, and finite unitary radiation without introducing a horizon-specific microscopic law.

The microscopic layer uses finite local Hilbert spaces, finite Weyl clock/shift pairs, partitioned reversible updates, frame/transport records, bounded load/capacity variables, and propagating matter and spin-2 registers. In the smooth sector, the flat BCC walk reproduces a Weyl/Dirac causal cone; geometry-conditioned propagation yields the ADM null cone; reciprocal controlled unitaries generate matter backreaction; and the gravitational kinetic sector is driven to the DeWitt trace coefficient $\lambda = 1/2$ by closure. Finite frame holonomies reconstruct three-dimensional curvature under refinement, and the reconstructed curvature tensor generates the physical transverse-traceless spin-2 Hessian. In the Schwarzschild linearized branch the odd-parity potential converges to the Regge-Wheeler form, while nondegenerate Darboux consistency selects the Zerilli partner.

The nonlinear covariance program is extended through full-six-component geometry and generic three-dimensional tests. Exact finite canonical commutators are impossible on a finite Hilbert space, so GCC does not claim a whole-Hilbert infinitesimal hypersurface-deformation algebra. Instead, finite Weyl/deformation maps and code/Ward identities are tested under spatial and register refinement. The resulting chain supports a controlled finite-quantum $\rightarrow$ discrete canonical gravity $\rightarrow$ ADM limit. In particular, the E45 Ward audit gives strong infrared H-H, H-D, and D-D closure while preserving the finite-dimensional no-go at the microscopic operator level.

Strong gravity is treated by a finite local capacity rather than by a singular continuum density. For astrophysical masses the outer trapping scale forms long before a Planck-density cap is reached. Smooth capped mass profiles can possess finite central curvature and an outer Schwarzschild-like horizon, but a density cap alone neither selects a bounce nor stabilizes a classical inner horizon. Inner-horizon surface gravity and mass-inflation audits instead drive the deep interior into a Planck-scale capacity regime. A spacelike finite-capacity transition placed inside the trapped region is causally compatible and removes the need to continue the infrared geometry to a long-lived Cauchy horizon. In rotation, the exterior is strongly driven toward Kerr-like infrared behavior by regularity, while microscopic Kerr uniqueness, the exact quadrupole relation, and the post-cap rotating geometry remain open.

The evaporation program reaches a constructive but deliberately limited endpoint. An eight-direction hard-core BCC radiation carrier streams exactly one nearest-neighbor edge per microscopic clock and is finite-frame covariant. A frame-covariant 8×4 Clifford matter carrier and an exactly energy-conserving matter-to-radiation vertex can be constructed on the same local Hilbert architecture. Their structural constraints reduce the large shell-mixing ambiguity to one scalar pulse angle θ . However, locality, unitarity, energy conservation, frame covariance, KMS detailed balance, and the existing capacity clock do not select that scalar. Treating θ as one infrared-matched datum reproduces the standard Schwarzschild mass scalings $E \propto M^{-1}$, $\dot{N} \propto M^{-1}$, $P \propto M^{-2}$, and $t_{\text{evap}} \propto M^3$, but the current single-gap carrier does not derive the full Hawking/greybody spectrum or a unique Page-time radiation correlator.

The final E56 audit finds no fatal internal contradiction after the required scope and normalization corrections. The strongest result is therefore not a claim of completed quantum gravity, but a coherent finite-QCA candidate with substantial constructive content and controlled infrared tests. The principal remaining gaps are concentrated in

microscopic theory selection: uniqueness of U_{GCC}^{full} , post-cap/source-supply dynamics, full species and greybody physics, and a distinctive quantitative observable derived without infrared calibration.

Keywords: quantum gravity; quantum cellular automaton; finite information; BCC lattice; finite Weyl algebra; ADM gravity; hypersurface-deformation algebra; Regge-Wheeler; Zerilli; regular black holes; Cauchy horizon; Kerr; Hawking radiation; finite-capacity quantum dynamics.

Claim-status summary

Fields: Element | Status | Meaning

- **Element:** Finite strict-local QCA — **Status:** Constructive — **Meaning:** Explicit finite reversible gate/scheduler prototypes exist; uniqueness is not established.
- **Element:** Relativistic matter cone — **Status:** Derived in the IR — **Meaning:** Flat and geometry-conditioned BCC sectors recover Weyl/Dirac principal propagation and a common causal cone in tested limits.
- **Element:** Einstein/ADM gravity — **Status:** Strong IR support — **Meaning:** DeWitt selection, curvature reconstruction, spin-2 dynamics, and refined HDA/Ward tests converge toward ADM/GR; exact whole-Hilbert HDA is not claimed.
- **Element:** Newton normalization — **Status:** Matched — **Meaning:** G is calibrated at the infrared interface rather than independently derived from the finite gate set.
- **Element:** Strong-gravity regularity — **Status:** Constructive/effective — **Meaning:** Finite-capacity regular-core profiles and causal transition geometries exist; the unique post-cap dynamics is open.
- **Element:** Rotation/Kerr — **Status:** IR selector / partial — **Meaning:** Frame dragging and horizon regularity strongly favor Kerr-like behavior; microscopic Kerr uniqueness and $q = -1$ are not derived.
- **Element:** Radiation carrier — **Status:** Constructive — **Meaning:** A strict-local frame-covariant BCC carrier and energy-conserving source vertex exist.
- **Element:** Evaporation coefficient — **Status:** Matched — **Meaning:** One scalar pulse angle remains after structural constraints and is fixed only by an infrared luminosity match in the minimal model.
- **Element:** Hawking spectrum/Page record — **Status:** Open — **Meaning:** A single characteristic-energy channel is not a thermal spectrum; source-supply, greybody, species, and unique Page-time correlations remain open.
- **Element:** Unique U_{GCC}^{full} — **Status:** Open — **Meaning:** Multiple finite unitary realizations share the tested infrared behavior.

1. Introduction

The central problem addressed by GCC is not simply how to quantize a classical metric. It is whether quantum matter, relativistic causal propagation, Einstein gravity, finite strong-gravity behavior, and finite black-hole radiation and information constraints can be understood as different sectors of one finite local update law. This integrated edition treats that question as its organizing principle.

The microscopic substrate is a fixed graph. The graph does not stretch, contract, or acquire continuum curvature as a fundamental object. What changes are finite records attached to nodes and Links: local frame/geometry information, matter amplitudes, capacity/load variables, and reversible scheduling data. Continuum distance, proper time, connection, metric, and curvature are relational observables reconstructed from these records. In this sense GCC differs in microscopic interpretation from a continuum theory in which the spacetime metric itself is fundamental, while still requiring general relativity as its smooth low-energy universality class.

The revision is built around a strict distinction between four logical levels. First, the microscopic QCA must be finite, unitary, autonomous, and causal. Second, a controlled smooth sector must recover relativistic matter and the ADM/DeWitt gravitational universality class. Third, spin-2 propagation and curvature must arise from the same finite frame/Link data rather than from an externally supplied wave equation. Fourth, strong-gravity and horizon statements must follow from the same finite records without a special horizon flag or a piecewise microscopic law. the integrated verification program also enforces a renormalization requirement: finite-register resolution must flow with lattice refinement whenever a continuum derivative would otherwise amplify cutoff errors.

The integrated logical map tested in this paper is

$$U_{GCC}^{full} \Rightarrow \{$$

finite Weyl/QCA matter and a common relativistic cone,
 ADM/Einstein gravity in a controlled infrared code sector,
 finite-capacity strong gravity without a local singular observable,
 Kerr-like rotating infrared behavior,
 finite unitary radiation transport with one matched scalar rate.

$$\}$$

The document is self-contained: the fixed-graph kinematics, local Hilbert factorization, matter and geometry sectors, curvature/spin-2 reconstruction, nonlinear covariance program, strong-gravity audits, rotation, radiation carrier, and final claim audit are all given below. The verification stack E1-E56 is used as an internal computational evidence hierarchy rather than as a replacement for analytic reasoning. Later negative controls revise several early candidate claims: in particular, a density cap does not imply a bounce, finite unitarity does not uniquely determine Hawking correlations, and the scalar evaporation strength is not fixed by the currently established microscopic symmetries.

PART I. ONE FINITE MICROSCOPIC LAW

2. Fixed graph, universal update, finite information, and capacity

GCC assumes a fixed relational BCC graph and a universal microscopic update label K . Coordinate labels and graph embeddings are auxiliary; only adjacency and relational records enter the microscopic law. The update label is an ordering variable, not directly an observable proper time. Local proper time and continuum lapse are emergent record-dependent quantities.

The reference microscopic scales are the Planck length and Planck time,

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}, \quad t_P = \frac{\ell_P}{c}.$$

For a resolved excitation that advances ΔN physical nearest-neighbor graph edges during ΔK microscopic clock updates, the grid-clock kinematic ratio is

$$\frac{v}{c} = \frac{\Delta N}{\Delta K},$$

provided one physical BCC nearest-neighbor edge is normalized to ℓ_P . Since the coordinate vectors are

$q_d = (\pm 1, \pm 1, \pm 1)$, this means one Cartesian coordinate unit corresponds to $\ell_P/\sqrt{3}$. One streaming hop per t_P then has physical speed c . This normalization becomes important in the final radiation-carrier construction.

Finite information is implemented by finite-dimensional local Hilbert spaces. Finite energy capacity is a distinct postulate: a four-component propagation carrier does not imply four independent Planck-energy slots. This separation corrects the earlier Minimal Action Saturation interpretation and prevents Hilbert-space dimension from being misread as local energy capacity.

$$ZX = \omega XZ, \quad \omega = e^{2\pi i/d}$$

Finite Weyl clock/shift pairs provide exact unitary record translations. Bounded Hermitian functions of these finite records are used for geometry, load, and local physical observables. The exact microscopic algebra is therefore finite; canonical commutators arise only in low-phase charts.

$$E_{cell} \leq E_P, \quad \rho \leq \rho_P$$

When one BCC processing node is identified with one GCC capacity cell at Planck energy and Planck density, the relational processing scale satisfies the following calibration.

$$4\ell_{proc}^3 = \ell_P^3, \quad \ell_{proc} = 4^{-1/3}\ell_P, \quad t_* = 4^{-1/3}t_P$$

This calibration fixes a branch of the microscopic scale dictionary. It does not turn the four propagation amplitudes into four independent energy capacities.

3. Full local Hilbert factorization

The minimal integrated architecture separates geometry, matter, capacity support, and reversible scheduling at nodes, and separates frame transport from scalar load on edges. This avoids the inconsistency that would result from using one Link phase simultaneously as a spin connection and as a strong-gravity load variable.

$$H = \otimes_x \left(H_{G_x} \otimes H_{M_x} \otimes H_{C_x} \otimes H_{W_x} \right) \otimes \otimes_e \left(H_{L_e}^{frame} \otimes H_{L_e}^{load} \right)$$

The geometry register G contains a finite inverse-triad-like record and its finite Weyl conjugate refresh. The matter register M contains the strict-local propagation carrier and any additional internal species records. The Compute register C carries local support/capacity information. The work register W stores reversible scheduler, memory, and temporary holonomy accumulation data. Each edge has a frame Link, which compares local frames and transports matter, and a load Link, which carries the finite-Weyl scalar mismatch used by the Compute-Link saturation sector.

4. Autonomous strict-local QCA

The fundamental law is a discrete-time unitary, not the exponential of a local Hamiltonian. A finite-time exponential of a nearest-neighbor Hamiltonian generically develops longer-range tails, whereas the microscopic GCC postulate requires a hard one-neighborhood causal cone per update.

$$|\Psi_{K+1}\rangle = U_{GCC}^{full} |\Psi_K\rangle$$

To combine several local tasks without allowing information to propagate through multiple freshly updated neighborhoods inside one K-step, the QCA is partitioned and double-buffered. An internal scheduler register makes the update autonomous.

$$U_{GCC}^{full} = \sum_{s=0}^{R-1} |s+1\rangle\langle s|_W \otimes \left(\prod_{B \in P_s} u_B \right)$$

Within each scheduler phase the blocks in the partition are disjoint. Every local gate reads records from the current slice and writes the next slice; freshly written records are not reused as controls until the next update. Consequently a local Heisenberg observable can spread by at most one graph neighborhood per microscopic step.

$$\text{supp}(U^\dagger O_x U) \subseteq A_{N_1(x)}$$

The scheduler is not an external clock program. Its phase is a finite internal record shifted by the same autonomous law. Several microscopic ticks may form a convenient Floquet supercycle for infrared analysis, but the fundamental law remains one repeated QCA update.

5. Two levels of exactness and microscopic gauge

A finite commutative site algebra admits no nontrivial ordinary Leibniz derivation. GCC therefore does not demand the continuum infinitesimal hypersurface-deformation algebra as an exact operator identity at a fixed finite cutoff. Instead it assigns exactness to finite matrix identities, gauge-covariant update composition, and finite groupoid/path consistency. The ADM/DeWitt algebra is an emergent smooth-sector requirement.

Tangential deformation on the fixed graph is interpreted as relational address relabeling rather than literal graph motion. The finite triad record also has an internal frame redundancy because the metric is invariant under a local rotation of the triad. These redundancies define the microscopic gauge sector. A candidate physical subspace is obtained by projecting or averaging over the frame and address gauge actions.

$$P_{frame} = \int dg U_{frame}(g), \quad P_{addr} = \frac{1}{|G_{addr}|} \sum_{\gamma \in G_{addr}} U_{addr}(\gamma)$$

$$P_{phys} = P_{addr} P_{frame}, \quad H_{phys}^{micro} = P_{phys} H, \quad \left[U_{GCC}^{full} P_{phys} \right] = 0$$

If the full update is gauge covariant, the physical projector commutes with the QCA and the restricted evolution remains unitary. A gauge-singlet uniform vacuum shows that the projected space need not be empty. This is a construction route, not a classification theorem for the complete 3+1 physical Hilbert space.

PART II. MATTER AND EMERGENT RELATIVITY

6. Uniform vacuum: the strict-local BCC/Weyl sector

In a uniform, low-load geometry the full update must reduce to a homogeneous propagation QCA. The exact flat reference remains the four-component doubled-Clifford BCC walk. The the E10-E31 stage integrated curved completion, however, uses a 32-dimensional direction-register dilation $C^8_{direction}$ tensor $C^4_{Clifford}$ as the universal local matter-carrier candidate. This distinction is forced by the verification audit: the 32-dimensional

family supports arbitrary sitewise geometry with one fixed BCC shift, whereas a naive pointwise splice between the exact four-component stencil and the curved 32-dimensional scattering rule is not unitary.

$$\alpha_i = \sigma_x \otimes \sigma_i, \quad P_i^\pm = \frac{I \pm \alpha_i}{2}, \quad A_s = P_x^{s_x} P_y^{s_y} P_z^{s_z}$$

$$\psi_{K+1}(n) = \sum_{s \in \{\pm 1\}^3} A_s \psi_K(n - s)$$

The update is a single radius-one convolution. It is not interpreted as three sequential physical shifts. In Fourier space the same candidate can be written as a product of finite Clifford rotations.

$$U(q) = e^{-iq_x \alpha_x} e^{-iq_y \alpha_y} e^{-iq_z \alpha_z}$$

After a fixed carrier-basis rotation, the four-component reference walk decomposes into two Weyl sectors of opposite chirality. Published classifications of homogeneous, local, isotropic two-component walks in three dimensions select the BCC Cayley geometry and the two Weyl walks within the stated class. GCC therefore retains the four-component walk as an exact flat benchmark. The verification update does not yet prove that this walk is an invariant four-dimensional subspace of the universal curved carrier; instead, the 32-dimensional family is required to reproduce the same Weyl principal cone in its low-wave-number flat sector.

7. Geometry-controlled propagation and the curved-space causal cone

The geometry record contains a finite triad-like variable. In a smooth semiclassical state it defines the spatial metric and inverse triad.

$$h_{ab} = E_a^i E_b^i, \quad h^{ab} = e_i^a e_i^b$$

The local lapse and shift are read from the same geometry/scheduler record family. The matter QCA is not curved by moving the graph. Instead, the local scattering/coin and Link transport are conditioned on the geometry record. Its principal velocity matrix is chosen to approach the following smooth expression.

$$V^a[G] = N^a I + cN e_i^a \alpha^i$$

Using the Clifford anticommutation relations, the local dispersion satisfies the ADM null-cone relation.

$$\left(\omega - N^a k_a\right)^2 = c^2 N^2 h^{ab} k_a k_b$$

Thus a fixed microscopic graph can support an emergent curved-spacetime causal cone. The frame Links supply the covariant transport connection, so the low-energy one-particle generator has the Weyl/Dirac principal form.

$$H_M^{IR}[G, L] = \frac{1}{2} \left\{ N^a I + cN e_i^a \alpha^i, -i\hbar D_a \right\} + H_{mass} + H_{int}$$

This establishes the kinematic curved-QFT branch conditionally: the exact finite QCA is fundamental, while the displayed differential operator is its smooth low-phase generator. In the E1 audit, a 3 x 3 x 3 inhomogeneous geometry pattern on the 32-dimensional carrier remained unitary with a relative residual 4.76 x 10⁻¹⁶, propagated to exactly the eight BCC neighbors in one tick, and recovered the flat Weyl eigenphase with an O(|k|³) absolute error. Interacting many-body QFT still requires additional local species and interaction gates.

8. Unitary matter backreaction from the same coupling

A separate classical source rule is unnecessary at the architectural level. Let a finite geometry basis control a matter unitary.

$$C_{GM} = \sum_g |g\rangle\langle g| \otimes U_M(g)$$

Because the conjugate geometry update is a finite shift, conjugating that shift by the same controlled matter unitary gives an exact finite-dimensional identity.

$$C_{GM}^\dagger (X_G \otimes I) C_{GM} = X_G \Omega_G, \quad \Omega_G = \sum_g |g\rangle\langle g| \otimes U_M(g + 1)^\dagger U_M(g)$$

In a low-phase chart, where the finite shift is represented by the exponential of a conjugate momentum and the matter update by a short effective Hamiltonian step, the identity reduces to the ordinary Hamiltonian backreaction force.

$$\Delta p_A = -t_* \frac{\partial H_M}{\partial q^A} + O(t_*^2)$$

The key point is reciprocity: geometry controls matter propagation, and the same unitary necessarily updates the conjugate geometry record according to the geometry dependence of the matter dynamics. The E2 verification extends this statement to a 32-dimensional anisotropic carrier with shear, lapse-scale, and nonzero shift: the finite geometry-shift identity has residual 4.13×10^{-17} , a one-site geometry change produces a one-site relative kick, and the unitary Hellmann-Feynman relation between the conjugate connection and the derivative of the low quasienergy agrees to better than 2×10^{-8} relatively in the tested branches. A complete proof of universal coupling across all matter species is still required.

PART III. EMERGENT EINSTEIN GRAVITY

9. Finite geometry dynamics

The geometry update has a bounded kinetic part, a curvature part derived from frame Links, and an address/shift part. In the smooth chart the effective Hamiltonian is required to approach the most general local quadratic DeWitt kinetic form with a scalar-curvature potential.

$$H_G[N, N^a] = \int d^3x \left\{ N \left[\frac{A}{\sqrt{h}} \left(\pi^{ab} \pi_{ab} - \lambda \pi^2 \right) - B \sqrt{h} R^{(3)} \right] + N^a D_a \right\}$$

At the microscopic level the momentum variables are bounded finite-Weyl functions rather than unbounded canonical operators. The displayed Hamiltonian is therefore an infrared generator of a finite unitary supercycle, not a fundamental continuous-time Hamiltonian. In the E4 reduced-sector construction, a bounded momentum function approaches the quadratic DeWitt kinetic term with an $\eta^{1.988}$ error law, reversible curvature compute-phase-uncompute returns the work register exactly to vacuum, and a three-phase autonomous Strang supercycle approaches its own bounded/cache effective generator with an $O(\tau^{2.020})$ error.

10. Closure selection of the DeWitt coefficient

The DeWitt trace coefficient is not left arbitrary. In the continuum bracket of two normal generators, the cross-term between the general quadratic kinetic term and the scalar-curvature potential produces the momentum constraint plus an additional trace-gradient term.

$$\{H_\lambda[N], H_\lambda[M]\} = D[\omega] + (1 - 2\lambda) \int d^3x \left(\nabla_a \pi \right) \left(N \nabla^a M - M \nabla^a N \right)$$

For generic states, if no additional trace constraint is introduced, the standard hypersurface-deformation algebra requires the coefficient of the extra term to vanish. Matching the normalization of the remaining momentum constraint also fixes the reciprocal kinetic/curvature product.

$$\lambda = \frac{1}{2}, \quad AB = 1$$

The resulting target infrared Hamiltonian density is the ADM gravitational Hamiltonian with an effective Newton constant.

$$H_G = \frac{16\pi G_{eff}}{\sqrt{h}} \left(\pi^{ab} \pi_{ab} - \frac{1}{2} \pi^2 \right) - \frac{\sqrt{h}}{16\pi G_{eff}} R^{(3)}$$

The selection is conditional on the assumed local quadratic kinetic sector, the scalar-curvature potential, and the absence of an additional trace constraint. Within that sector it sharpens the earlier numerical observation that the closure root converges to the general-relativistic value.

11. Frame-Link holonomy and emergent spatial curvature

The fixed graph does not acquire microscopic curvature. Curvature is reconstructed from non-integrable frame transport. For each oriented BCC edge, the frame Link transforms covariantly under local frame rotations.

$$L_{x,s} \rightarrow G_x L_{x,s} G_{x+s}^\dagger, \quad L_{x+s,-s} = L_{x,s}^\dagger$$

A minimal BCC four-edge loop provides a finite holonomy.

$$W_{x;t} = L_{x,s} L_{x+s,t} L_{x+t,s}^\dagger L_{x,t}^\dagger$$

In a smooth low-phase chart, the loop holonomy measures the curvature two-form through its area bivector.

$$W_{x;s,t} = \exp \left[-\frac{i}{4} \Sigma_{s,t}^{ab} F_{ab}^i \sigma_i + O(\ell_{proc}^3) \right]$$

Independent BCC loop bivectors span the three independent spatial two-form directions. Contracting the recovered frame curvature with the inverse triad gives the scalar curvature required by the ADM potential.

$$R^{(3)} = \epsilon_{ij}^k e_i^a e_j^b F_{ab}^k$$

A subtle locality point is essential. A node cannot instantaneously read all four Links of a loop whose opposite corner lies two graph steps away. GCC therefore uses a reversible work register to accumulate the loop product over several microscopic ticks. The E3-E4 verification makes this constructive: a controlled load-to-frame source gate is exactly unitary, a non-Abelian loop angle begins quadratically when the underlying Link angle is linear in load, the calibrated holonomy reproduces $f_g(\chi)$ to about 1.6×10^{-12} across a finite load register, and a four-tick loop product agrees with the direct holonomy to about 8×10^{-17} . The completed cache is then uncomputed at the supercycle boundary, avoiding a permanent work-mode degree of freedom.

the integrated verification program extends this construction from a scalar-curvature target to the full smooth frame-curvature tensor. General-triad BCC clovers recover the curvature two-form and directional Ricci data with approximately second-order error. On a Schwarzschild isotropic slice the scalar trace tends to zero while $R_{\text{rhat_rhat}}$ remains nonzero, and contracting the reconstructed Riemann/Ricci tensor with the physical sheet-TT representation yields the spin-2 curvature Hessian directly. A fixed 2T frame alphabet alone is too coarse for weak transport; the successful finite architecture separates the non-Abelian gauge label from a near-identity transport record whose effective resolution improves under block refinement.

12. Infrared hypersurface-deformation algebra

The geometric origin of the H-H structure function can be seen directly from the response of a unit normal to a normal deformation. Orthogonality to the tangents gives an inverse-metric-weighted lapse gradient.

$$\delta_N n = -h^{ab} (\partial_b N) e_a$$

The order difference of two normal deformations is therefore tangential and has the standard inverse-metric structure.

$$[\delta_N, \delta_M] X = e_a h^{ab} (N \partial_b M - M \partial_b N)$$

At the canonical level this is the target normal-normal bracket.

$$\{H[N], H[M]\} = D \left[h^{ab} (N \partial_b M - M \partial_b N) \right]$$

The existing numerical audits support the distinction between full-cutoff and smooth-sector closure. The earlier low-mode results below remain useful as a regulator audit. Version 3.1 established a full three-dimensional canonical test with six h_{ab} and six π^{ab} components in which bounded finite-phase momentum and curvature maps are inserted into the V4 Hamiltonian. With no deformation, H-H, H-D, and D-D residuals refine approximately as $a^{1.946}$, $a^{1.927}$, and $a^{1.983}$. If the finite-register deformation parameters are refined jointly as $\eta_p = \zeta_R = 0.30$ a, the fitted powers are $a^{1.969}$, $a^{2.680}$, and $a^{1.983}$, respectively. The DeWitt scan continues to select $\lambda = 1/2$.

Fields: epsilon | Raw H-H diamond | Residual after fitted shift | Fitted coefficient

- **epsilon:** 0.040 — **Raw H-H diamond:** 3.17×10^{-4} — **Residual after fitted shift:** 2.52×10^{-6} — **Fitted coefficient:** 0.99552
- **epsilon:** 0.020 — **Raw H-H diamond:** 7.93×10^{-5} — **Residual after fitted shift:** 3.01×10^{-7} — **Fitted coefficient:** 0.99723
- **epsilon:** 0.010 — **Raw H-H diamond:** 1.98×10^{-5} — **Residual after fitted shift:** 3.49×10^{-8} — **Fitted coefficient:** 0.99810
- **epsilon:** 0.005 — **Raw H-H diamond:** 4.96×10^{-6} — **Residual after fitted shift:** 4.16×10^{-9} — **Fitted coefficient:** 0.99853

Fields: Sites | a | Low-mode H-D residual | Low-mode D-D residual

- **Sites:** 32 — **a:** 0.19635 — **Low-mode H-D residual:** 0.11249 — **Low-mode D-D residual:** 0.17904
- **Sites:** 64 — **a:** 0.09817 — **Low-mode H-D residual:** 0.02875 — **Low-mode D-D residual:** 0.04623
- **Sites:** 128 — **a:** 0.04909 — **Low-mode H-D residual:** 0.007227 — **Low-mode D-D residual:** 0.01165

- **Sites:** 256 — **a:** 0.02454 — **Low-mode H-D residual:** 0.001809 — **Low-mode D-D residual:** 0.002918

A crucial negative control accompanies this result. If the bounded deformation is held fixed while only the spatial lattice is refined, the H-H residual does not converge to the ADM target; at N=40 the tested fixed-deformation branch remains near 2.88×10^{-2} . Thus finite-register corrections must themselves flow under refinement. The E6 block construction below supplies a concrete collective-mode mechanism for that flow, but these results are still evidence for the required infrared universality class rather than a proof that every nonlinear many-body sector of the full QCA is anomaly-free.

the integrated verification program repeats the closure test after matter and two physical TT sectors are integrated. With a common inverse-metric structure function, the H-H error again converges approximately as dx^2 , while a 10 percent TT gradient-normalization mismatch leaves a finite closure floor. Finite H[N] and D[xi] circuit families then reproduce physical-sector H-H group commutators and geometry-aware H-D/D-D commutators under joint spatial, geometry-register, and smearing-register refinement. The full-state H-H circuit remains open because the current normal update does not yet evolve the complete geometry canonical pair.

13. Einstein-Hilbert reconstruction and normalization status

If the geometry, matter, gauge, and physical spin-2 sectors described above share the same smooth lapse, shift, and inverse-metric structure function, if the complete geometry-inclusive H-H/H-D/D-D algebra closes in the refined physical sector, and if no additional low-energy gravitational degree of freedom survives, the effective action takes the Einstein-Hilbert plus matter form with higher-derivative ultraviolet corrections. the integrated verification program strengthens this route by constructing the smooth directional-curvature and spin-2 sectors, an all-register finite-unitary existence prototype, and explicit smeared deformation circuits. The remaining covariance gap is no longer hidden in the matter or TT sector: it is localized to the full geometry canonical action in the normal-deformation family, while extra frame-mode decoupling remains a separate dynamical requirement.

$$S_{eff} = \frac{c^3}{16\pi G_{eff}} \int d^4x \sqrt{-g} R + S_{matter} + \Delta S_{UV}$$

The relative normalization between the DeWitt kinetic term and curvature term is constrained by closure. The absolute value of the effective Newton constant is not independently derived in the present construction and remains matched to the GCC Planck calibration. The E6 block audit gives an explicit finite-Weyl collective flow $\eta_{eff}=\eta/B$ and $\zeta_{eff}=\zeta/B$, showing how bounded microscopic corrections can weaken under coarse graining, but it does not yet establish the stability or uniqueness of the absolute Newton normalization in the full interacting theory.

PART IV. FINITE CAPACITY AND REGULAR STRONG GRAVITY

14. Capacity saturation and robust core scaling

No elementary capacity cell is required to represent arbitrarily large localized energy. The density bound alone implies a minimum volume for a fixed enclosed mass. In spherical symmetry this gives the robust cube-root core scale.

$$r_c = \left(\frac{3M}{4\pi\rho_p} \right)^{1/3} \propto M^{1/3}$$

This scaling does not require a Hayward metric, a particular Link harmonic, or an equal Compute-Link load assumption. It follows from finite density, three-dimensional volume counting, and finite total mass. The detailed radial profile requires an additional constitutive closure.

15. Explicit Compute-Link circuit and the correlated radial equation

The earlier quadratic strong-gravity branch can be realized by two required local support roles: Compute and Link. The revised architecture retains the explicit reversible correlation circuit rather than replacing its quantum correlations by an arbitrary radial interpolation. Using the mean enclosed density as the coarse load gives the dimensionless variable below.

$$\chi(r) = \frac{3m(r)}{4\pi\rho_p r^3}$$

After tracing only reversible work registers, the explicit Compute-Link circuit gives the normalized density response

$$f_g(\chi) = \frac{\rho}{\rho_p} = \chi^2 + \chi(1 - \chi)\sin^2(g\chi^2)$$

The response is bounded and has exact vacuum and saturation endpoints.

$$0 \leq f_g(\chi) \leq 1, \quad f_g(0) = 0, \quad f_g(1) = 1$$

Combining mass conservation with the definition of the mean-density load yields a closed radial equation for the full correlated circuit.

$$r \frac{d\chi}{dr} = -3\chi(1 - \chi)\cos^2(g\chi^2)$$

This equation improves the logical status of the strong-gravity branch. The radial dependence is no longer freely chosen once the circuit response is specified. The remaining completion choice is the microscopic identification between the scalar Link load and the coarse spherical load.

16. Hayward limit, correlated deformations, and exterior behavior

In the uncorrelated limit the radial equation integrates exactly.

$$g = 0: \quad \chi(r) = \frac{a^3}{r^3 + a^3}, \quad m(r) = \frac{Mr^3}{r^3 + a^3}, \quad a^3 = \frac{3M}{4\pi\rho_p}$$

This is the Hayward-type mass profile used in the earlier revision. For nonzero circuit correlation the exact Hayward form is deformed, but the density cap and central saturation remain intact. Near the vacuum the correlation correction enters only at fifth order in the load; therefore the leading $p=2$ exterior correction remains quartic in inverse radius, while the new correlation-dependent correction appears at much higher inverse power. The detailed coefficient is microscopic-model dependent.

At the center the load approaches unity and the density approaches the Planck value. The corresponding static spherical geometry is de Sitter-like rather than singular. This establishes a regular strong-gravity branch conditionally on the chosen local constitutive dictionary; it does not prove dynamical collapse into that branch or stability of an inner horizon.

17. Frame curvature and scalar load must be distinct Link factors

A key consistency correction is that the frame-holonomy phase of Section 11 cannot be identified directly with the scalar finite-Weyl load phase. In the weak-field Einstein sector, curvature is proportional to energy density; in the minimal two-channel branch the density is quadratic in the load. Therefore the curvature holonomy begins quadratically in the scalar load, not linearly.

$$H_L = H_{L^{frame}} \otimes H_{L^{load}}$$

$$\theta_{curvature} \propto f_g(\chi) = \chi^2 + O(\chi^5)$$

The two Link factors remain parts of one neighboring-record architecture. The frame factor transports local orientation and reconstructs curvature; the scalar factor carries the bounded capacity mismatch. In the E3 source-gate prototype the load register controls two noncommuting $SU(2)$ frame rotations; their loop holonomy is quadratic at weak load and is calibrated once so that the normalized curvature response equals $f_g(\chi)$ from vacuum to saturation. The controlled source gate is unitary to machine precision and its reciprocal load-shift identity is exact. This makes the separation constructive while leaving the unique microscopic source gate and the full three-dimensional frame-to-curvature dictionary open.

PART V. HORIZONS, INFORMATION, AND FINITE RADIATION

18. Horizons are emergent causal cuts, not microscopic defects

The full QCA contains no horizon flag. Event-horizon membership is a global property of the emergent causal structure, while the local curvature at the outer horizon of a very massive black hole can be arbitrarily small. A horizon-specific microscopic gate would therefore require nonlocal information about the global state. GCC instead uses the same local update everywhere. An outer horizon is identified only after a semiclassical state is coarse-grained into causal inside/outside sectors.

The same principle constrains information claims. Finite local Hilbert spaces and exact reversibility prohibit literal many-to-one erasure in a closed QCA, but they do not determine a unique black-hole-to-radiation unitary. Later E50-E55 negative controls demonstrate that Page-like entropy turnover, nonzero radiation correlations, and discrete

entropy are all compatible with many inequivalent finite unitaries. The correct present claim is therefore **finite-unitary compatibility**, not an exact unique information-transfer law.

19. Boundary algebra, area bookkeeping, and what remains conditional

The four-component flat carrier still provides a useful kinematic inside/outside split. For a unit spatial normal $\hat{n}$, define

$$P_{out} = \frac{I + \alpha_{\hat{n}}}{2}, \quad P_{in} = \frac{I - \alpha_{\hat{n}}}{2}, \quad \text{rank } P_{out} = \text{rank } P_{in} = 2.$$

The outgoing support algebra is therefore isomorphic to a two-dimensional carrier algebra,

$$A_H = P_{out} M_4(C) P_{out} \simeq M_2(C).$$

This is an exact kinematic statement about the reference carrier. It is not a proof that a complete many-body horizon possesses exactly one binary memory unit per elementary support, nor that its microscopic entropy spectrum is equally spaced. If one imposes a minimal binary boundary-support model, the area increment $\Delta A = 4 \ln 2 \ell_p^2$ follows only after matching to the semiclassical Bekenstein-Hawking normalization. The E50/E51 audits show that such counting by itself does not select a unique emission spectrum.

20. Finite local observables and the end of classical mass inflation

Let O be a bounded local observable on a finite local Hilbert space and let one exact QCA step be U . Unitary conjugation preserves its norm,

$$O_{K+1} = U^\dagger O_K U, \quad \|O_{K+1}\| = \|O_K\|.$$

Therefore finite-clock differences satisfy

$$\|\Delta O\| \leq 2 \|O\|, \quad \|\Delta^2 O\| \leq 4 \|O\|.$$

If a strong-gravity curvature/capacity observable has a finite spectrum on the allowed nondegenerate local register, the microscopic QCA cannot realize a literal infinite local curvature or an infinite update in a finite clock step. This does **not** prove a stable inner horizon. Rather, the E47 analysis shows that the classical mass-inflation branch reaches the finite-capacity regime rapidly and must then leave its continuum domain of validity.

21. Spacelike capacity transition and information preservation

A continuous static regular-center metric with $f(0) > 0$, $f(\infty) > 0$, and a trapped band $f < 0$ generically requires at least two zero crossings. Thus a static regular black hole tends to inherit an inner horizon. The E47 mass-inflation audit shows why this is dynamically problematic. GCC therefore treats the static two-horizon metric as an effective compatibility model, not as the unique microscopic endpoint.

Inside a trapped band, a constant-radius transition surface can be spacelike. The infrared spacetime can consequently terminate on a finite-capacity hypersurface before the classical Cauchy horizon is reached. The future of this surface is carried by the finite QCA rather than by a continuation of the classical metric. Reversibility excludes a closed-system absorbing singular sink, but it does not by itself choose between every possible reversible post-cap branch. In particular, a black-to-white-hole bounce remains allowed in principle but is **not derived** by the density cap or by unitarity.

22. Finite-frame radiation carrier and local emission vertex

The final constructive radiation architecture uses eight hard-core directional rails associated with the BCC vectors $q_d = (\pm 1, \pm 1, \pm 1)$. If $n_d(x, K) \in \{0, 1\}$ is the occupation of rail d at site x , strict-local streaming is

$$n_d(x, K + 1) = n_d(x - q_d, K).$$

This map is a basis permutation and therefore defines an exact unitary on arbitrary superpositions. With the physical nearest-neighbor BCC edge normalized to ℓ_p , one hop per t_p propagates at speed c .

To repair the anisotropy of the early E12 matter interface while retaining the same $8 \times 4 = 32$ local register size, a frame-covariant BCC Clifford carrier can be constructed,

$$H_{fc} = \frac{\kappa}{\sqrt{3}} \sum_{i=1}^3 D_i \otimes \gamma_i,$$

where D_i contains the Cartesian components of the eight BCC directions. A fourth Clifford matrix Γ_4 anticommutes with the three spatial γ_i and is a scalar under spatial frame rotations. Let P_p and P_n denote the positive- and negative-energy projectors. The direction-resolved lowering map is

$$J_d = P_{n,d} \Gamma_4 P_{p,d}.$$

Let C_d be the hard-core creation operator on outgoing rail d , and let $A_d = C_d^\dagger$ be the corresponding annihilation operator. The local source vertex is

$$H_{emit} = \sum_{d=0}^7 \left(J_d \otimes C_d + J_d^\dagger \otimes A_d \right), \quad A_d = C_d^\dagger.$$

Assigning radiation energy equal to the matter gap gives

$$\left[H_{emit}, H_{matter} + H_{rad} \right] = 0$$

to numerical precision in the exact finite tests. The vertex is finite-frame covariant and direction preserving; hard-core blocking is automatic when the matching rail is occupied. The only continuous freedom is the scalar pulse angle in

$$U_{emit}(\theta) = e^{-i\theta H_{emit}}.$$

23. Thermality, matched rate, and the present information limit

The E55 audit proves that the remaining scalar θ is not fixed by the currently established structural principles. The continuous family $U_{emit}(\theta)$ preserves the same unitarity, locality class, total energy, and finite-frame covariance for all θ . KMS detailed balance fixes only a ratio,

$$\frac{\Gamma_\uparrow}{\Gamma_\downarrow} = e^{-\beta \Delta E},$$

while an overall rate remains free. The existing capacity clock likewise supplies only coarse trigger fractions and does not determine the required weak emission probability.

If spectral continuity with the isotropic E12 carrier is imposed, the E54 normalization is

$$\kappa = \sqrt{0.45^2 + 0.35^2 + 0.25^2}, \quad \Delta = 2\kappa = 1.244989959799.$$

For a local transition occurring at proper distance ρ outside a large Schwarzschild horizon, the near-horizon redshift gives

$$\frac{E_\infty}{k_B T_H} = 2\pi \Delta \frac{\rho}{\ell_p}.$$

At the effective core length $\rho = L_{core} = \sqrt{3/(8\pi)} \ell_p$, this is $E_\infty \simeq 2.70263 k_B T_H$, numerically close to the mean energy of an ideal massless bosonic number spectrum. The coincidence is a **characteristic-energy scale**, not a derivation of a thermal spectrum: the present single-gap carrier has zero intrinsic spectral width and no derived greybody factor.

If the one remaining scalar is matched to the standard Schwarzschild luminosity coefficient, the minimal model requires

$$p = \sin^2 \theta = \frac{1}{960\pi \Delta \lambda^2}, \quad \lambda = \frac{L_{core}}{\ell_p},$$

which gives $p \simeq 2.23116 \times 10^{-3}$ and $\theta \simeq 4.72528 \times 10^{-2}$ rad. One local Planck-scale gate plus redshift then yields

$$E_\infty \propto M^{-1}, \quad \dot{N} \propto M^{-1}, \quad P \propto M^{-2}, \quad t_{evap} \propto M^3.$$

The exponents are structural; the coefficient is matched. A one-shot source has an exact geometric first-emission waiting-time distribution,

$$P(T = n) = (1 - p)^{n-1} p,$$

but a complete Page-time record requires a source-supply/re-excitation law, inter-source entanglement, multiple energies/species, and exterior scattering. Those remain open.

PART VI. INTEGRATED AUDIT AND VERIFICATION PROGRAM

24. Evidence hierarchy and program completion

The integrated paper uses five distinct evidential labels. **Constructive/exact finite** means that a finite gate, permutation, matrix identity, or reversible scheduler has been built explicitly. **Derived infrared** means that a controlled spatial/register/code limit approaches a continuum target. **Selector/compatibility** means that regularity, causality, or a boundary condition narrows a branch without deriving the complete microscopic law. **Matched** means that a coefficient is fixed by a known infrared normalization rather than predicted independently. **Open** means that the current QCA architecture does not determine the quantity.

The verification stack E1-E56 is complete as a research program. Program completion is not theory completion: the final microscopic U_{GCC}^{full} is not unique, several infrared coefficients remain matched, and no unique accessible quantitative astrophysical discriminator has yet been derived.

25. Final claim-status matrix

- **Finite strict-local QCA:** constructive. Explicit reversible local gate/scheduler prototypes exist; uniqueness is open.
- **Relativistic matter and common cone:** derived in tested infrared sectors. The original E12 matter block is an interface; E54 supplies a frame-covariant completion candidate of the same 8×4 register size.
- **Einstein/ADM gravity:** strong infrared support. DeWitt selection, curvature reconstruction, spin-2 dynamics, and E39-E45 HDA/Ward refinement approach the ADM target. Exact whole-Hilbert infinitesimal HDA is not claimed.
- **Newton normalization:** matched. G is fixed at the infrared/Planck calibration interface rather than derived independently from the finite gate set.
- **Strong gravity:** constructive/effective. Finite-capacity regular profiles and a spacelike capacity transition are compatible with an ordinary outer horizon. A density cap does not force a bounce, and the unique post-cap geometry is open.
- **Rotation/Kerr:** infrared selector/partial. Weak frame dragging and horizon regularity strongly favor Kerr-like exterior behavior. Microscopic $q = -1$, exact extremality, and the unique rotating post-cap branch are not derived.
- **Radiation carrier:** constructive. E54 gives an actual nearest-neighbor BCC carrier and an exactly energy-conserving, finite-frame-covariant local source vertex.
- **Evaporation rate:** matched. Structural constraints reduce the emission freedom to one scalar pulse θ , but locality, unitarity, energy conservation, frame covariance, KMS detailed balance, and the existing capacity clock do not determine it.
- **Hawking spectrum/Page record:** open. The single-gap channel supplies a characteristic energy scale and one-source waiting-time law, not a full thermal/greybody spectrum or a unique Page-time correlator.
- **Unique fundamental law:** open. E51 and E55 exhibit inequivalent finite unitary families compatible with the currently imposed structural principles.

26. Cross-stage consistency corrections

The final E56 audit finds no fatal internal contradiction after the following scope corrections.

- E45 supersedes the E31 stopping point at the **infrared/code Ward** level; it does not create an exact finite-dimensional whole-Hilbert HDA.
- The E48 sub-grid inner scale belongs to the spherical effective E46 profile; Kerr r_- is macroscopic for ordinary astrophysical spin.
- The spacelike capacity transition is not a derived black-to-white-hole bounce.
- Kerr-like quadrupole behavior is an infrared regularity selector, not a microscopic derivation of the exact Kerr relation.
- The early E12 32-dimensional matter law is a constructive interface rather than the final frame-covariant physical matter Hamiltonian; E54b is a completion candidate.
- Spectral continuity fixes the E54 carrier normalization to the E12 isotropic gap without introducing an additional energy scale.

- The scalar evaporation pulse θ and all coefficient-normalized luminosities based on it must be labeled **infrared matched**.
- The value $E_\infty \simeq 2.70263 k_B T_H$ is a characteristic-energy scale, not a derivation of a thermal spectrum.
- Hard-core radiation rails are a finite-QCA carrier/regulator and do not by themselves derive bosonic or fermionic species statistics.
- One physical BCC nearest-neighbor edge is normalized to ℓ_p ; equivalently one Cartesian coordinate unit in $q_d = (\pm 1, \pm 1, \pm 1)$ is $\ell_p/\sqrt{3}$. One streaming hop per t_p therefore has speed c .

27. Falsifiability and non-predictions

GCC has meaningful mathematical and structural falsifiability. A proposed microscopic gate can fail strict locality or unitarity; the final carrier can fail to preserve a common causal cone; the spatial/register refinement can fail to approach ADM/HDA; or a proposed strong-gravity completion can require an unbounded local observable despite finite registers. These are genuine tests and several E-series negative controls were designed to expose precisely such failures.

Empirical discrimination is currently weaker. The present construction does **not** robustly predict a black-to-white-hole bounce, observable gravitational-wave echoes, an independently normalized Hawking luminosity, a unique thermal/greybody Hawking spectrum, or a unique Page-time radiation correlator. Planck-core corrections to ordinary astrophysical ringdown are negligible in the tested branch. The highest-value future discriminator is therefore a source-supply/radiation correlation law derived uniquely from the final U_{GCC}^{full} , because E50-E51 show that generic unitarity and Page-like entropy turnover are not distinctive.

28. Verification endpoint and remaining tasks

The verification program can be grouped into six completed blocks:

- **E1-E13:** fixed-graph matter, reciprocal backreaction, bounded geometry, gauge/frame structure, curvature reconstruction, and universality tests.
- **E14-E31:** strong-gravity observability controls, rotation, finite TT propagation, Regge-Wheeler/Zerilli reconstruction, all-register finite-unitary integration, and smeared-deformation circuits.
- **E32-E45:** full-six nonlinear geometry, generic three-dimensional curvature, finite-Hilbert no-go analysis, two-stage quantum-to-canonical-to-ADM refinement, and final canonical Ward closure.
- **E46-E49:** horizon-before-cap ordering, regular-core compatibility, inner-horizon mass inflation, spacelike Cauchy-horizon replacement, and rotating/Kerr strong-gravity audits.
- **E50-E55:** observable-discriminator negative controls, microscopic emission non-uniqueness, frame-covariant radiation carrier, exact source vertex, scalar-coupling no-go, and infrared-matched minimal evaporation bookkeeping.
- **E56:** global consistency, claim hierarchy, falsifiability, and final maturity audit.

The planned verification program is therefore complete. The remaining scientific work is not another routine verification stage but theory selection and phenomenology: select or derive a sufficiently unique U_{GCC}^{full} , construct the macroscopic source-supply/re-excitation law, derive multi-energy/species and greybody dynamics, and obtain a distinctive quantitative observable without importing its normalization from the infrared target theory.

PART VII. EXTENDED VERIFICATION: CURVATURE, SPIN-2 DYNAMICS, AND SMEARED DEFORMATIONS

29. Finite frame curvature, renormalization, and universality (E10-E13)

E10 replaces the earlier scalar-holonomy sketch by a constructive smooth-sector curvature reconstruction. Four BCC directions generate six independent direction pairs whose area bivectors span the spatial two-form space. Path-ordered frame Links and four-edge clovers are reconstructed at a common basepoint, and least-squares inversion recovers the frame curvature two-form. In conformally flat tests the scalar-curvature relative error falls from 6.82×10^{-2} at $a=0.08$ to 1.10×10^{-3} at $a=0.01$ with fitted power 1.988. A general non-diagonal triad gives the same approximately second-order behavior. Local $SO(3)$ frame rotations preserve the scalar to roundoff.

A finite-representation negative control is essential. A fixed binary-tetrahedral 2T/T frame alphabet has a minimum nontrivial rotation of 120 degrees and therefore cannot by itself represent arbitrarily weak smooth Links. The successful finite representation is a product architecture: a finite non-Abelian endpoint gauge label together with a finite near-identity transport record. The transport resolution must refine faster than the area scale. In the constructive block map, a b^3 block averages the collective transport increment as $1/B$, giving an effective link-angle resolution proportional to a^4 ; the integrated finite-frame test then recovers general-triad scalar curvature with fitted power about 2.03. This is a constructive RG mechanism, not a uniqueness theorem for the final frame representation.

E11-E13 examine whether the enlarged representation leaves unwanted low-energy modes. Global transport shifts are pure gauge for difference-only Link sources, the BCC incidence operator has one exact collective null direction, and relative/frame excitation gaps separate parametrically from the collective mode in the tested blocks. The universality audit distinguishes relevant data - causal-cone speed and anisotropy, matter species/gaps, the Newton/weak-load normalization, DeWitt/HDA structure, and any ungapped frame modes - from $O(a^2 k^3)$ lattice corrections that vanish in the IR. Strong-load constitutive functions can agree in the weak sector while remaining physically different near saturation; therefore the IR universality class does not by itself select a unique fundamental cellular automaton.

30. Strong-gravity observability, angular momentum, and spin-squared response (E14-E18)

E14-E15 separate robust strong-gravity statements from observational overreach. For a Planck-scale regular core in a Hayward-type diagnostic exterior, ordinary shadow and eikonal-ringdown shifts scale as powers of the core-to-horizon ratio and are negligible for astrophysical masses. A near-horizon reflector would produce echo times in an observable band, but such a reflector is not part of the present GCC horizon construction; echoes therefore remain conditional. The finite density cap fixes a cube-root minimum core scale but does not choose bounce, stall, or mass-flux dynamics. Once an outer horizon has formed, strict locality prevents the saturated core from sending a new classical signal through that horizon. These results disfavor treating the core itself as the generic source of observable merger-wave modifications.

Rotation likewise forces the geometry sector beyond a pointlike saturated core. A diagnostic bound $J \leq M R c$ shows that an astrophysical dimensionless spin cannot be stored solely in a Planck-sized core. E16 constructs exact finite angular-momentum exchange between matter and Link/frame records; on a periodic BCC graph the global vector record is conserved phase by phase, and the weak frame-dragging field converges to the transverse $J \times r/r^3$ form with $O(a^2)$ anisotropy error. E17 matches the asymptotic Kerr frame-drag coefficient $\kappa=2$ as an IR normalization, but lower-order mass and spin data do not determine the spin-squared quadrupole amplitude. E18 makes the STF spin-squared source finite and reciprocal; standard slow-rotation horizon regularity supplies an infrared Kerr selector, while the microscopic normalization and nonlinear rotating branch remain open.

31. Unitary no-hair and propagating finite TT radiation (E19-E22)

E19 demonstrates that local no-hair relaxation is compatible with exact finite unitarity. A localized anomalous quadrupole can transfer amplitude to outward frame/wave records under a time-independent nearest-neighbor unitarity. On finite rings recurrence is unavoidable; before the return time the local survival probability decays while total norm is preserved. A spectral audit shows the precise failure mode: a stable bound eigenmode with finite defect overlap produces persistent hair, whereas a purely dispersive continuum permits local relaxation.

E20 then applies this criterion to the actual microscopic architecture and finds an important omission: the E12 frame scaffold has no vacuum inter-Link propagation gate, so an abstract radiation continuum cannot honestly be identified with the E12 spectrum. E21 fills this gap. Each physical TT polarization receives a finite Q/P qudit, onsite kinetic phases, and eight strict-local BCC matching-gradient phases. The linearized supercycle obeys the BCC graph-wave Floquet law and approaches $\omega=c|k|$ with $O(a^2)$ spatial-plus-splitting error at fixed CFL ratio. Exact finite-qudit diagonalization shows that the low quasienergy gaps converge to the same oscillator-like spectrum.

E22 removes an unnecessary phenomenological degree of freedom introduced by the earlier bound-mode diagnostic. In the minimal actual architecture, the spin-squared source is linear in the same TT field; it shifts the static equilibrium but does not create an independent defect oscillator. Completing the square gives a source-independent perturbation Hessian. A wrong quadrupole coefficient is therefore a deformation of the propagating TT field and can dephase into the wave sector. The remaining static freedom is the source normalization/strong-background equilibrium, not a separately postulated local no-hair oscillator.

32. Curved spin-2 sectors, finite constraint reduction, and the Schwarzschild chain (E23-E27)

E23 establishes that curved transport alone is not enough. Replacing the flat E21 gradient by a minimally coupled Schwarzschild scalar-like operator leaves a finite strong-field mismatch from the $l=2$ Zerilli potential under radial refinement. The missing information is the gauge/constraint reduction of the metric perturbation. Once the reduced Zerilli master operator is supplied, however, the same finite Q/P architecture carries it: nearest-neighbor radial gradient phases plus an onsite local curvature phase are exact finite unitaries, and the lowest radial spectrum converges at approximately second order. On the Schwarzschild branch the Zerilli coefficient is an algebraic function of the local lapse/load and areal-radius data, so no nonlocal mass label is required [19].

E24 constructs the finite reduction mechanism explicitly. Four finite canonical pairs can be mixed by an exact finite symplectic map, constrained and gauge-fixed to a two-dimensional phase space, and represented quantum mechanically by a rank- d physical projector inside d^4 configuration states. A radius-one centered stencil reconstructs the Zerilli-Moncrief coordinate with $O(h^2)$ error. Because the coordinate contains one derivative of a metric register, its finite resolution must scale one power faster: $\Delta K=O(h^3)$ is required to retain an $O(h^2)$ master-variable error, while algebraic registers may scale as $O(h^2)$. A nondynamical auxiliary constraint variable generates the missing even-parity onsite term through a local Schur complement, and the same elimination has an exact finite-field projector analogue.

E25-E27 then identify the physical curvature data. On the standard time-symmetric Schwarzschild slice $R^{(3)}=0$, but the orthonormal Ricci eigenvalues are nonzero: $R_{\hat{r}\hat{r}}=-2M/r^3$ and the two tangential eigenvalues are $+M/r^3$. The even constraint coefficient can therefore be written locally in terms of lapse and directional curvature. Locality, weak-field matching, endpoint behavior, and stability still allow a continuous family of strong-curvature corrections. The ambiguity is removed in the verified Schwarzschild linear sector by parity consistency. The covariant spin-2 curvature action on the sheet-TT tensor gives $3 R_{\hat{r}\hat{r}}$, so the odd potential is

$$U_{\text{odd}} = f\left[\frac{l(l+1)}{r^2} + 3R_{\hat{r}\hat{r}}\right] = f\left[\frac{l(l+1)}{r^2} - \frac{6M}{r^3}\right].$$

This is the Regge-Wheeler form [20]. Direct BCC holonomy reconstruction on the isotropic Schwarzschild slice drives the inferred $l=2$ coefficient to $k_{\text{eff}}=3.99963$ at $a=0.05$, with approximately $O(a^2)$ convergence. Requiring the independently derived odd potential and a minimal local rational even constraint sector to form a nondegenerate Darboux pair then selects

$$C_2(y) = \frac{14y(3y+8)}{(3y+4)^2}, \quad y = \frac{2M}{r}.$$

and the generated even potential agrees symbolically with the Zerilli potential. The same construction extends to all $l \geq 2$ and is consistent with the standard Regge-Wheeler/Zerilli Darboux relation [21]. This is a strong infrared selection chain rather than a derivation of the full nonlinear GCC constraint algebra.

33. Microscopic spin-2 curvature Hessian (E28)

E28 closes one more gap between continuum geometry and the finite wave carrier. The original E12 frame/source gate contains finite geometry, work, transport, and gauge-frame records but no independent TT amplitude pair, so a spin-2 Hessian cannot be obtained by Taylor expanding E12 alone. The DeWitt kinetic form resolves this cleanly: on the physical trace-free plus/cross plane, $\text{Tr}(\pi_{\text{TT}})=0$ and the generalized DeWitt metric reduces to a positive identity independent of the trace coefficient. The two E21 Q/P carriers are therefore the minimal physical kinetic extension.

More importantly, the factor 3 in the Schwarzschild spin-2 curvature term is not stored as a fitted constant. The full BCC-reconstructed orthonormal Riemann and Ricci tensors are contracted directly with the sheet-TT representation. The resulting two-by-two curvature Hessian becomes polarization independent and its mean eigenvalue divided by $R_{\hat{r}\hat{r}}$ converges to 3 with fitted power 2.001. At $a=0.05$ the ratio is 3.00122 and the plus/cross eigenvalue split is at roundoff. A rotation of the TT basis transforms the Hessian covariantly to about 2.5×10^{-17} .

The reconstructed Hessian then controls an exact finite two-polarization Q/P onsite phase. With 21 states per polarization the local Hilbert sector has dimension 441; the unitary residual is about 1.8×10^{-13} , the first-excitation subspace capture exceeds 0.9999, and the local stiffness converges to the Regge-Wheeler value with $O(a^2)$ holonomy error. An autonomous 11-phase TT scheduler combines kinetic, BCC gradient, and curvature phases with exact inverse and zero scheduler leakage. The result is a constructive finite spin-2 gate, not a proof that this extension is the unique microscopic completion.

34. All-register finite-unitary existence and non-uniqueness (E29)

E29 assembles the previously separate finite sectors into one factorized 18-phase supercycle. The phases perform geometry-controlled matter evolution, reversible work computation, load/frame/transport backreaction, work uncomputation, capacity refresh, reversible loop-curvature caching, TT half-kinetic evolution, eight BCC matching-gradient layers, a curvature-controlled spin-2 phase, the second half-kinetic layer, and curvature-cache uncomputation. The formal local-patch Hilbert dimension is about 9.04×10^{13} , so a dense matrix is neither necessary nor practical; every factor is instead represented as a finite controlled unitary or permutation.

On an exact 32×7^4 superposed matter-plus-TT basis sector, the maximum norm drift through the full cycle is 2.8×10^{-15} , the work and curvature caches return to vacuum, and the explicit inverse recovers the initial amplitudes to about 5.0×10^{-15} together with all discrete control records. Four coherently superposed geometry/capacity/frame/loop sectors also return under the full inverse. The non-Abelian frame projector commutes with the source evolution to roundoff, and the original matter-backreaction identity survives tensor extension by the TT sector.

The curvature cache supplies a direct local bridge from finite transport Links to the spin-2 phase. For an oriented four-edge Z_7 loop, vertex gauge shifts cancel exactly in the loop sum; exhaustive and structured sweeps show no invariance or cache-inverse failures. Nevertheless E29 does not establish uniqueness. Two different symmetric splittings of the same finite Q/P Hamiltonian are exact unitaries, differ microscopically by $O(dt^3)$, and share the same $O(dt^2)$ continuum quasienergy. The correct the E10-E31 stage claim is therefore existence and compatibility of an all-register finite update, not a uniquely selected fundamental rule.

35. Integrated HDA and finite smeared deformation circuits (E30-E31)

E30 first tests the integrated canonical target after matter and spin-2 fields have been added. Matter, TT+, and TTx Hamiltonians close on one total diffeomorphism constraint when they share the same inverse-metric structure function $\beta = h^{xx}$; the centered-difference H-H error scales as $dx^{1.99}$. Arbitrary ultralocal matter and curvature-controlled TT potentials cancel from the H-H bracket to roundoff. A 10 percent TT propagation-coefficient mismatch instead leaves an approximately 8.5 percent closure floor, showing that a common causal/metric normalization is a nontrivial integrated condition.

The H-D test exposes a second necessary condition. If the geometry is frozen while $D[xi]$ moves only matter/TT fields, the closure error approaches an approximately 8 percent floor. Adding the Lie drag of the spatial metric restores H-D convergence with fitted power about 1.99; D-D converges similarly. Joint finite-geometry refinement with $\Delta h = O(dx^2)$ preserves the full effective target, while fixed Δh leaves H-H/H-D floors. These results extend the earlier E5 renormalization lesson to the matter-plus-spin-2 system.

E31 then replaces effective Poisson evaluation by actual finite smeared circuit families. For smooth finite-dimensional canonical fields define $S_H[N, \epsilon] = \exp(\epsilon A_H[N])$ and $S_D[xi, \epsilon] = \exp(\epsilon A_D[xi])$. With ϵ proportional to dx^2 , the H-H group commutator approaches the diffeomorphism circuit on the matter and two TT sectors; the combined circuit error falls from 0.284 at $N=32$ to 5.97×10^{-3} at $N=256$. When the metric participates in the tangential circuit, H-D and D-D commutator errors scale approximately as dx^2 . Finite lapse/shift alphabets are also compatible provided their resolution is derivative-aware, $\Delta \text{smear} = O(dx^3)$; a fixed alphabet fails to approximate arbitrary continuum smearings under refinement.

The decisive negative result is the full-state H-H circuit. Two present normal-deformation circuits leave h unchanged, while the target diffeomorphism produced by their commutator Lie-drags h . The physical matter/TT error converges, but the full-state relative error remains about 78.8 percent in the finest tested case, almost entirely carried by the missing metric deformation. This is not treated as a numerical anomaly. It identifies the required microscopic register: a canonical geometry momentum/extrinsic-curvature sector. E31d constructs a finite diagonal DeWitt shear over F_{31} , $q \rightarrow q+n$ G_p with $G = I - (1/2)11^T$, and verifies exact symplecticity and 536,238 forward/inverse state-lapse combinations without failure. A complete full H-H proof must still add the curvature momentum shear and the full frame/geometry diffeomorphism action.

36. Interim conclusion through E31

The E10-E31 block establishes that finite local frame/transport records can reconstruct directional curvature under refinement, that physical TT registers can propagate on the same BCC graph, and that the reconstructed curvature tensor can generate the local spin-2 Hessian rather than importing a continuum graviton equation by hand. The Schwarzschild linearized chain reaches Regge-Wheeler and, through nondegenerate Darboux parity consistency, the Zerilli partner. An 18-phase factorized all-register finite-unitary prototype also exists and survives coherent-control inverse tests.

The principal E31 stopping point was the complete geometry-inclusive normal deformation: matter/TT H-H and geometry-aware H-D/D-D circuits showed the correct tendency, but frozen or incomplete geometry momentum produced a residual error floor. The next verification block therefore enlarged the geometry register, tested generic 3D curvature, and replaced impossible exact finite CCR requirements by finite Weyl/code/Ward closure.

PART VIII. NONLINEAR FULL-GEOMETRY AND CANONICAL HDA (E32-E45)

37. Full-six nonlinear geometry and three-dimensional curvature (E32-E35)

E32 replaces reduced geometry charts by all six independent components of a symmetric spatial metric and their DeWitt momenta. The finite register is organized so that the quadratic kinetic form approaches

$$G_{abcd} \pi^{ab} \pi^{cd} = \pi^{ab} \pi_{ab} - \frac{1}{2} \pi^2$$

in the smooth chart. The spin-2 subspace extracted from the full register reproduces the earlier TT spectrum, while nonlinear negative controls show where the linearized construction ceases to be exact.

E33 moves to a nonlinear qutrit metric chart and tests the canonical response of the bounded ADM kinetic gate. Finite Weyl packets recover the expected semiclassical directional derivatives; geometry and matter can be included in one reversible supercycle; and finite-frame covariance survives the nonlinear chart. E34 then evaluates nonlinear H-H, H-D, and D-D closure in a reduced plane-symmetric ADM sector. The classical/discrete target converges approximately quadratically in the spatial spacing, while operator-valued structure functions can be represented by finite circuits. The important negative control is that an onsite kinetic-only model gives a trivial H-H bracket and therefore cannot represent gravity.

E35 restores genuinely three-dimensional curvature. A finite nonlinear Ricci stencil containing face and mixed-derivative channels converges near second order. Deleting the mixed derivatives produces a large nonvanishing error floor, demonstrating that the full tensor structure is not optional. The same local curvature can be encoded either by the metric stencil or by BCC frame holonomies with the expected $O(a^2)$ matching in the tested smooth backgrounds.

38. Finite-Hilbert no-go and the correct closure target (E36-E38)

A finite-dimensional Hilbert space cannot support an exact canonical commutator $[Q, P] = i\hbar I$ because the trace of any commutator vanishes. GCC therefore rejects an exact infinitesimal whole-Hilbert hypersurface-deformation algebra as the microscopic target. The exact finite objects are Weyl translations and exponentiated deformation maps,

$$ZX = \omega XZ, \quad U_D(\xi) = e^{-iD[\xi]}, \quad U_H(N) = e^{-iH[N]},$$

with the continuum algebra required only on an appropriate low-phase/code sector under refinement.

E36-E37 test this distinction directly. Finite Weyl registers reproduce canonical response increasingly well as the register dimension grows, while exponentiated commutator circuits show that an arbitrary low-dimensional projection does not automatically recover the desired deformation composition. E38 demonstrates a second obstruction: insufficient spatial resolution can dominate the apparent algebra even when the local register is adequate. Correct canonical cell momentum normalization and joint spatial/register refinement are therefore necessary.

39. Two-stage quantum-to-ADM convergence (E39-E40)

E39 introduces a local-overlap engine that evaluates finite quantum HDA kernels without materializing an exponentially large global Hilbert matrix. The quantum result is first compared with the **same discrete classical** canonical system, and only then is that discrete classical system compared with the continuum ADM target. This two-stage comparison prevents register error from being confused with spatial discretization error.

For the refined classical sequence $N = 16 \dots 128$, representative H-H, H-D, and D-D errors converge approximately as $a^{1.9} - a^{2.0}$. At $N = 32, d = 5$, the quantum-to-same-discrete H-H comparison gives a best-fit amplitude near unity and cosine similarity above 0.998. The remaining H-D error is larger but decreases under register refinement. The central conclusion is the controlled chain

$$\text{finite quantum} \rightarrow \text{finite/discrete canonical gravity} \rightarrow \text{ADM/GR}.$$

E40 performs the spatial and register limits jointly at fixed physical chart. At $N = 128$, $d = 5$, the H-H and D-D shape errors are at the 10^{-3} level and the H-D error is a few percent; additional ensembles show the result is not a single-smearing accident. Finite-Weyl packet refinement simultaneously drives the canonical commutator error from order unity at very small d to below 10^{-7} in the tested high- d packet sector.

40. Generic 3D universality and factorized full-six closure (E41-E44)

E41 extends the classical HDA audit to all six metric components, a scalar matter sector, and a full three-component shift. Directional classical H-H, H-D, and D-D errors again show near-second-order trends. Matter and spin-2 sectors can be tuned to the same infrared cone, and the integrated finite-frame tests remain compatible with this common propagation speed.

E42 tests the ingredients needed for a generic three-dimensional full-six quantum calculation: three-axis structure-function covariance, scalar-matter H-H universality, full-six quantum spin-2 propagation, and the nonlinear mixed-derivative curvature stencil. The mixed-derivative negative control leaves an order-tens-of-percent residual, confirming that the successful result is not generated by a reduced diagonal stencil.

E43 then factorizes the generic-3D H-H calculation. For a Hamiltonian $H_i = T_i + V_i$ with commuting onsite kinetic terms and diagonal curvature potentials,

$$[H_i, H_j] = [T_i, V_j] - [T_j, V_i],$$

so the required local commutator can be evaluated without a $d^{6 \times 27}$ global matrix. Register refinement gives a strong convergence sequence: the direct local H-H shape error falls from order 10^{-1} at $d = 3$ to below 10^{-3} at $d = 7$. Removing mixed-edge curvature channels again produces a large error, while the scalar-matter H-H kernel converges to the same discrete target.

E44 adds direct D-D and H-D tests. The finite quantum D generator correctly Lie-drags both metric and momentum when compared with the **correct discrete conservative target** rather than a continuum product-rule split. The remaining nonlinear direct H-D convergence is slower, motivating the Ward formulation of E45.

41. Canonical Ward closure (E45)

E45 is the strongest present HDA result. Instead of demanding an impossible exact finite infinitesimal operator identity, it combines the exact finite-Weyl canonical response of the D generator with the nonlinear directional derivative of the Hamiltonian. The H-D Ward relation converges rapidly with register dimension: representative shape error falls from order unity at $d = 3$ to $\sim 2.7 \times 10^{-2}$ at $d = 5$, $\sim 1.9 \times 10^{-3}$ at $d = 9$, and below 10^{-6} at $d = 15$ in the tested local sector. D-D shows a similar trend.

Thirty random smooth lapse/shift ensembles at fixed coarse spatial lattice preserve the result. At $d = 9$, representative H-D and D-D shape errors are of order 10^{-5} and 10^{-4} respectively. The correct statement is therefore

strong IR canonical HDA closure

with the following explicit limitation:

no exact whole – Hilbert nonlinear operator HDA is claimed.

PART IX. STRONG GRAVITY, INNER HORIZONS, AND ROTATION (E46-E49)

42. Horizon-before-cap ordering and regular-core compatibility (E46)

Take the Planck density

$$\rho_p = \frac{c^5}{\hbar G^2}.$$

For a homogeneous spherical mass M , the radius at which the mean density reaches the cap is

$$R_{cap} = \left(\frac{3M}{4\pi\rho_p} \right)^{1/3},$$

while the Schwarzschild trapping scale is

$$R_H = \frac{2GM}{c^2}.$$

The equality $R_H = R_{cap}$ occurs at $M \simeq 0.173 m_p$ in this simple ordering model. For astrophysical masses the ratio grows as $M^{2/3}$ and becomes enormous. Thus a Planck-density modification can occur deep inside an already-formed outer horizon without changing the ordinary exterior horizon formation.

A useful effective compatibility profile is

$$m(r) = M \frac{r^3}{r^3 + r_0^3}, \quad r_0^3 = \frac{3M}{4\pi\rho_p},$$

with

$$f(r) = 1 - \frac{2Gm(r)}{c^2 r}.$$

Near $r = 0$,

$$f(r) = 1 - \frac{r^2}{L_{core}^2} + O(r^5), \quad L_{core}^2 = \frac{3c^2}{8\pi G\rho_p}.$$

For the Planck cap, $L_{core} \simeq 0.34549 \ell_p$. The effective profile has finite central curvature and, above a threshold, an outer Schwarzschild-like horizon plus an inner horizon. This is **not** a derived GCC equation of state or unique mass profile. It is a demonstration that finite density, finite curvature, and an ordinary outer horizon are mutually compatible.

A separate collapse chronology audit using the general-relativistic Oppenheimer-Snyder/free-fall benchmark up to the cap confirms that horizon crossing precedes cap activation. The post-cap branch is not supplied by general relativity and must be determined by the finite QCA. A negative control constructs static, bouncing, and accreting capped branches that all satisfy $\rho \leq \rho_p$; therefore the density cap alone does not imply a bounce.

43. Inner-horizon instability and the finite-capacity regime (E47)

For the regular effective profile, the inner-horizon surface gravity is nonzero except at the degenerate horizon-pair threshold. In the astrophysical large-mass limit the inner scale approaches a mass-independent Planck value and the e-fold time associated with the classical blueshift becomes Planckian. A schematic perturbation amplitude

$$A(y) = \epsilon \frac{e^y}{(1+y)^p}, \quad y = \kappa_- v,$$

shows the structural problem: for any finite p , exponential blueshift ultimately defeats a power-law tail. Even extremely small seeds reach a nonlinear/capacity regime after only $O(10-100) t_p$ in the effective spherical branch.

The important negative result is that a bounded state amplitude is not enough to guarantee bounded derivatives. A finite QCA regularity claim must involve both a bounded local spectrum and a bounded finite update. This is precisely what the exact finite-unitary norm bounds of Section 20 supply. The conclusion is not “the inner horizon is stable,” but rather

classical mass inflation $\rightarrow$ capacity scale $\rightarrow$ breakdown of the infrared continuum branch.

44. Replacing the Cauchy-horizon continuation (E48)

A continuous static regular-center metric with a trapped band generically has an even number of horizon crossings. Random smooth regular mass-profile tests confirm the elementary intermediate-value argument: if $f(0) > 0$, $f(\infty) > 0$, and $f < 0$ somewhere in between, at least two zeros occur. Thus removing the inner horizon while keeping a static regular center is not generic.

In ingoing Eddington-Finkelstein coordinates,

$$ds^2 = -f(r)c^2 dv^2 + 2c dv dr + r^2 d\Omega^2,$$

outgoing radial null curves satisfy

$$\frac{dr}{dv} = \frac{c}{2} f(r).$$

Inside the trapped band $f < 0$, a constant- r transition surface is spacelike. Placing a finite-capacity surface at $r_{sat} > r_-$ intercepts outgoing rays at finite advanced time instead of allowing them to asymptote indefinitely to the classical Cauchy horizon. Closed-system unitarity excludes a many-to-one absorbing singular sink, but multiple reversible future-core branches remain possible. The minimal current interpretation is therefore a spacelike transition into finite unitary QCA dynamics, not a derived white-hole bounce.

45. Rotating strong gravity and the Kerr infrared branch (E49)

Rotation changes the inner-horizon scale substantially. For the Kerr benchmark,

$$r_{\pm} = r_g \left(1 \pm \sqrt{1 - \chi^2} \right), \quad r_g = \frac{GM}{c^2},$$

so r_- is macroscopic for ordinary astrophysical spin. The spherical E48 statement $r_- < \ell_p$ is therefore profile-specific and must not be transferred to Kerr. What remains microscopic in the rotating case is the final capacity layer generated by the exponentially amplified inner-horizon perturbation.

A second structural result concerns angular momentum. If all astrophysical J were localized as ordinary matter motion inside a Planck core, the diagnostic ratio

$$\eta_J = \frac{J}{McL_{core}}$$

would be enormous. The rotating branch therefore requires extended frame/Link degrees of freedom to carry angular momentum, consistent with the weak frame-dragging construction.

Outer-horizon regularity strongly suppresses the anomalous slow-rotation quadrupole sector. Evaluated at a Planck stretched horizon, the coefficient multiplying the horizon-singular Hartle-Thorne branch must be suppressed by a factor of order ℓ_p/M . This makes a Kerr-like quadrupole an excellent infrared selector, but it is not a microscopic derivation of $Q = -J^2/M$ from U_{GCC}^{full} . The exact extremality relation is likewise not fixed by the density cap alone.

PART X. OBSERVABILITY AND FINITE RADIATION (E50-E55)

46. Observable-discriminator audit (E50)

The first task is negative: remove signatures that are attractive but not actually predicted. Planck-core corrections to ordinary light-ring/ringdown observables scale as a high power of ℓ_p/M and are fantastically small for astrophysical black holes. A deep core inside an ordinary horizon does not create a robust echo unless an additional near-horizon reflector is introduced. Horizon regularity drives the exterior quadrupole toward Kerr rather than toward an observable non-Kerr deformation. Finite-entropy corrections become appreciable only near the Planck-mass regime.

Unitary evaporation itself is also not distinctive. Random finite pure states can have nearly identical Page-like entropy curves while distributing pairwise mutual information very differently. Likewise, finite entropy does not select a unique line spacing: choices such as $\Delta S = \ln 2$ and $\Delta S = \ln 3$ are both finite but predict different frequencies. Strict locality, finite memory, and exact unitarity still admit inequivalent emitted-record correlation matrices. The correct discriminator must therefore be derived from the actual microscopic emission/source-supply law, not inferred from generic unitarity.

47. Microscopic horizon-to-radiation channel and its non-uniqueness (E51)

For a finite core energy register e and outgoing energy register r , exact energy conservation requires the local emission unitary to be block diagonal in shells of fixed $q = e + r$. If shell q has dimension d_q , the allowed unitary is $U(d_q)$; even a real time-reversal-symmetric choice leaves $d_q(d_q - 1)/2$ continuous angles. In the E51 toy register, hundreds of real mixing angles remain accessible from radiation-vacuum input. Thus

$$\text{energy conservation} + \text{locality} + \text{reversibility} \not\Rightarrow \text{unique emission amplitudes.}$$

A constructive Householder shell mixer demonstrates that a finite, energy-conserving, exactly reversible emission gate exists, and it can be embedded as a nineteenth phase of the E29 scheduler without breaking reversibility. But an alternative gate with the same structural properties gives a radically different spectrum and correlation matrix. This converts “evaporation ambiguity” from a qualitative caveat into a direct mathematical non-uniqueness result.

48. Internal transition selector from the existing matter law (E52)

The E12 32-dimensional matter label is not a 32-level energy ladder. It decomposes as

$$C^{32} = C_{direction}^8 \otimes C_{Clifford'}^4$$

its computational basis is not an energy basis, and the geometry-conditioned Hamiltonians do not all commute. Moreover, the original E29 eighteen-phase scheduler acts as $U_{E29} \otimes I_R$ on an appended radiation register: there is no hidden emission channel to reinterpret.

Nevertheless, the existing anisotropy operator contains a distinguished **internal** transition tensor. At the isotropic point,

$$A = P_n H_1 P_p.$$

has rank 16 and equal nonzero singular values. Within the operator span $aH_0 + bH_1$, every energy-changing block is proportional to A . This reduces the relative transition ambiguity sharply, but it does not yet provide an outgoing quantum mode or a frame-covariant physical matter Hamiltonian. That missing physical carrier is supplied constructively in E54.

49. Frame/Link routing and redshift calibration (E53)

The existing TT sector is a propagating spin-2 mode, not a generic Hawking matter carrier, and its finite free spectrum is not exactly resonant with the E52 matter gap. The old E12 32-dimensional interface is also not fully frame covariant under genuine BCC spatial rotations because of its unequal fixed coefficients. However, the BCC direction labels themselves supply a parameter-free angular router: the same finite frame rotation permutes matter and outgoing direction channels, and the router commutes exactly with the twelve-element finite rotation group used in the frame audit.

Near-horizon redshift produces the correct Hawking mass scaling independently of the scalar emission rate. For a local gap ΔE_p at proper distance ρ ,

$$E_\infty \simeq \alpha \Delta E_p, \quad \alpha \simeq \frac{\rho}{2r_s},$$

and since

$$k_B T_H = \frac{\hbar c}{4\pi r_s},$$

one obtains

$$\frac{E_\infty}{k_B T_H} = 2\pi \Delta \frac{\rho}{\ell_p}.$$

The black-hole mass cancels. If one local attempt occurs per local Planck time, the redshift also slows the external attempt rate by another factor α , so the power scales as $\alpha^2 \propto M^{-2}$.

50. Frame-covariant physical radiation carrier (E54)

E54 closes the carrier-level gap constructively. Eight hard-core BCC rails give an exact finite streaming QCA. A frame-covariant 8×4 Clifford carrier with

$$H_{fc} = \frac{\kappa}{\sqrt{3}} \sum_i D_i \otimes \gamma_i$$

has a spectrum $\pm \kappa$ with rank 16 per sign sector and is invariant under the tested finite frame rotations. A scalar Clifford matrix Γ_4 supplies an exact energy-flip isometry. Combining this matter transition with the eight outgoing rails gives the local source vertex of Section 22.

The vertex and streaming map form one reversible end-to-end channel: a quantum is created at the source, travels one nearest-neighbor BCC edge per clock, and the entire sequence can be inverted to machine precision. The full local Fock register has dimension $32 \times 2^8 = 8192$; the interaction is Hermitian, energy conserving, frame covariant, and automatically hard-core blocked on an occupied rail. This is the strongest constructive evaporation result of the present paper.

51. Scalar coupling, characteristic spectrum, and source count (E55)

Spectral continuity with the isotropic E12 interface fixes the otherwise arbitrary E54 normalization,

$$\kappa = \sqrt{0.45^2 + 0.35^2 + 0.25^2},$$

without introducing a second scale. The remaining scalar pulse θ , however, survives every structural audit. KMS detailed balance fixes only $\Gamma_\uparrow/\Gamma_\downarrow$; finite unitarity, energy conservation, frame covariance, and the capacity-clock orbit all permit a continuous family of θ values.

If θ is treated as one infrared-matched datum, the minimal model is fully normalized at the level of characteristic energy and luminosity. The present $2.70263 k_B T_H$ gap lies within 5.4×10^{-4} relatively of the mean energy of an ideal massless bosonic number spectrum, but the single-gap line has zero width while the thermal distribution has $O(1)$ width. The result is therefore a characteristic scale, not a thermal-spectrum derivation.

For one source, the first-emission waiting time is geometric. Since a two-sector source can emit only one gap quantum before ending in the lower sector, macroscopic evaporation requires many source degrees or a re-excitation/exposure law. Entropy bookkeeping is compatible with such a surface picture: if each characteristic quantum removes $x_0 = E/(k_B T_H)$ entropy units, then

$$\frac{N_{emit}}{A/\ell_p^2} = \frac{1}{4x_0}.$$

For $x_0 = 2.70263$, this is approximately one characteristic emission event per 10.81 Planck-area cells. This is a kinematic compatibility result, not a derived activation law.

PART XI. FINAL CLAIM AUDIT, FALSIFIABILITY, AND CONCLUSION (E56)

52. Global consistency and claim hierarchy

E56 classifies every major statement by logical strength. The hierarchy used in this final edition is:

1. **Exact finite / constructive:** a finite matrix identity, local gate, reversible scheduler, or explicit finite model has been built and checked.
2. **Derived infrared:** a controlled low-phase/spatial/register limit approaches the stated continuum structure.
3. **Selector / compatibility:** a consistency or boundary condition singles out a branch within a stated class, without deriving the complete microscopic law.
4. **Matched:** a dimensionful or dimensionless coefficient is fixed using a known infrared normalization rather than derived independently.
5. **Conditional/effective:** a useful model demonstrates compatibility but is not uniquely selected by the microscopic architecture.
6. **Open:** the present construction does not determine the quantity.

The cross-stage audit finds no fatal contradiction after ten necessary clarifications. The most important are: E45 is an IR Ward closure rather than a whole-Hilbert operator identity; the E48 sub-grid inner horizon belongs only to the spherical effective profile; the spacelike capacity transition is not a derived bounce; Kerr $q = -1$ is an IR regularity selector; the old E12 matter block is an interface rather than the final physical carrier; the E55 scalar is matched; and the physical BCC nearest-neighbor edge, not each Cartesian component of q_d , is normalized to ℓ_p .

53. Falsifiability and the absence of a unique present discriminator

GCC has nontrivial mathematical falsifiability: the finite gates can fail to be unitary or local, the refinement sequence can fail to approach ADM/HDA, the common cone can split, or the strong-gravity finite register can fail to support a bounded local observable. These are genuine structural tests and many negative controls in the E-series were designed precisely to expose such failures.

Empirical falsifiability is presently weaker. The current construction does **not** robustly predict a bounce, observable echoes, an independently normalized Hawking luminosity, a full thermal/greybody spectrum, or a unique Page-time correlator. Ordinary astrophysical ringdown deviations from a Planck-scale core are too small in the present branch. The strongest future discriminator would be a radiation/source-supply correlation law derived uniquely from the final U_{GCC}^{full} , because E50-E51 demonstrate that generic Page curves and generic unitary correlations are nonunique.

54. Final discussion and conclusion

The verification program has reached a useful endpoint. GCC now possesses explicit finite constructions for strict-local matter propagation, reciprocal backreaction, bounded geometry dynamics, frame/Link curvature, a

physical spin-2 sector, an all-register finite-unitary prototype, nonlinear full-six geometry registers, refined canonical HDA/Ward closure, finite-capacity strong gravity, rotating Kerr-like infrared behavior, a spacelike Cauchy-horizon replacement, and a frame-covariant radiation carrier with an exactly energy-conserving local source vertex.

The remaining gaps are concentrated rather than diffuse. First, the present principles do not select a unique fundamental U_{GCC}^{full} . Second, the post-cap/source-supply dynamics that would turn many one-shot local emitters into a complete evaporation history is not derived. Third, the final matter/species spectrum, greybody scattering, and a unique radiation correlator remain open. Fourth, several infrared coefficients - most importantly Newton normalization, the weak frame-dragging normalization, and the scalar evaporation rate in the minimal channel - are matched rather than independently predicted.

These limitations determine the scientifically appropriate conclusion. GCC should be described as an **advanced finite-QCA quantum-gravity candidate framework with substantial constructive and controlled-infrared support**. It should not be described as a completed or unique quantum theory of gravity. The significance of the present result is that a large fraction of the usual continuum structure can be carried by finite local records and finite reversible updates while the remaining unsolved questions are sharply identified and experimentally or mathematically distinguishable in principle.

APPENDIX A. Exact finite-Weyl and BCC audit definitions

For a finite Weyl pair, matrix exactness can be monitored by the relative Frobenius residual

$$R_W = \frac{\|W_M W_N - \omega^k W_N W_M\|_F}{\|W_M W_N\|_F}$$

Fields: d | Weyl phase exponent | Relative matrix residual

- **d: 5 — Weyl phase exponent: 2 — Relative matrix residual: 2.18×10^{-16}**
- **d: 7 — Weyl phase exponent: 4 — Relative matrix residual: 5.75×10^{-16}**
- **d: 11 — Weyl phase exponent: 8 — Relative matrix residual: 3.54×10^{-16}**
- **d: 17 — Weyl phase exponent: 14 — Relative matrix residual: 4.31×10^{-16}**

The reported values are floating-point roundoff, illustrating the distinction between exact finite matrix composition and emergent continuum closure.

APPENDIX B. Backreaction identity

Let the finite geometry shift advance the geometry basis by one record and let the matter update be diagonal in the geometry-control basis. Direct multiplication gives

$$C_{GM}^\dagger (X_G \otimes I) C_{GM} = X_G \Omega_G, \quad \Omega_G = \sum_g |g\rangle\langle g| \otimes U_M(g + 1)^\dagger U_M(g)$$

No measurement or classical stress extraction is used. In a smooth chart, expand the neighboring controlled unitaries to first order in the geometry-record step and identify the shift phase with the conjugate geometry momentum. The resulting momentum kick is the derivative of the same matter Hamiltonian that controls propagation. Higher-order Baker-Campbell-Hausdorff terms are ultraviolet corrections rather than independent source terms.

APPENDIX C. DeWitt one-half selection

For the general local quadratic kinetic term, only the kinetic-curvature cross bracket contributes second derivatives of the lapses. Contracting the curvature variation with the generalized momentum combination gives a standard momentum-constraint term and a trace-gradient remainder proportional to one minus twice the DeWitt coefficient. Antisymmetrizing the lapse pair and integrating by parts yields the expression in Section 10. If no additional trace constraint is admitted, standard HDA closure therefore selects one half. This is a conditional derivation within the local quadratic ansatz, not a global uniqueness theorem over arbitrary nonlocal or higher-derivative microscopic completions.

$$\{H_\lambda[N], H_\lambda[M]\} = D[\omega] + (1 - 2\lambda) \int d^3x \left(\nabla_a \pi \right) \left(N \nabla^a M - M \nabla^a N \right)$$

APPENDIX D. BCC loop curvature and strict locality

The BCC neighbor shell contains independent pairs of edge directions whose cross products span the spatial bivector space. Four-edge loops therefore provide enough independent holonomies to reconstruct the three spatial curvature two-form components in the smooth sector. A direct one-tick plaquette readout would violate radius-one causality because the opposite loop corner is not in the node neighborhood. The reversible scheduler instead transports and multiplies one edge factor per microscopic tick. A completed loop record is cached only after all factors have arrived. The finite delay is derivative suppressed in the infrared and becomes a genuine ultraviolet memory effect near the processing scale.

$$W_{x;s,t} = L_{x,s} L_{x+s,t} L_{x+t,s}^\dagger L_{x,t}^\dagger$$

$$W_{x;s,t} = \exp\left[-\frac{i}{4}\sum_{s,t}^{ab} F_{ab}^i \sigma_i + O(\ell_{proc}^3)\right]$$

APPENDIX E. Correlated strong-gravity radial equation

Using the spherical mean-density load, mass conservation and the explicit Compute-Link circuit imply

$$r \frac{d\chi}{dr} = -3\chi(1 - \chi) \cos^2(g\chi^2)$$

The uncorrelated limit is logistic and gives the analytic Hayward-type branch. For finite correlation the equation is separable in principle but not generally elementary. The endpoints remain fixed because the circuit correction vanishes at zero and unit load. The weak-load expansion preserves the quadratic leading term, while the correlation correction starts at fifth order. Hence the central density cap and cube-root mass scaling are more robust than the detailed exterior coefficient.

APPENDIX F. Physical-state construction at microscopic level

The microscopic gauge problem is separated from the emergent Hamiltonian constraint problem. Internal frame rotations and address relabelings act as finite redundancies of the record description. Projecting onto their invariant sector gives a candidate microscopic physical Hilbert space.

$$P_{phys} = P_{addr} P_{frame'}, \quad H_{phys}^{micro} = P_{phys} H, \quad \left[U_{GCC}^{full} P_{phys} \right] = 0$$

A nonzero invariant vacuum guarantees that the projected space is not empty. The E8 Z3 plaquette model and E9 binary-tetrahedral 2T model admit explicit gauge projectors, gauge-invariant curvature observables, and finite evolution that preserves the projected sector. the integrated verification program adds a gauge-invariant finite loop-curvature cache and shows that the projected frame sector remains compatible with the all-register TT extension. The actual GCC frame/address representation remains to be selected, and physical frame excitations must still be shown to decouple dynamically. What has changed is the curvature status: a smooth inhomogeneous frame-holonomy-to-curvature reconstruction is now constructive in the tested backgrounds, although its uniqueness and full nonlinear embedding are not yet established.

APPENDIX G. Finite radiation carrier and information-preserving source vertex

The final radiation construction uses eight hard-core rails per BCC site. The local radiation Hilbert dimension is $2^8 = 256$. Streaming is the strict-local permutation

$$n_d(x, K + 1) = n_d(x - q_d, K).$$

The frame-covariant matter carrier has dimension $8 \times 4 = 32$. Let $P_{p,d}$ and $P_{n,d}$ denote the positive- and negative-energy projectors in direction sector d , and let Γ_4 be the spatial-frame scalar Clifford flip. The direction-resolved lowering map is

$$J_d = P_{n,d} \Gamma_4 P_{p,d'}$$

The full local interaction on the hard-core Fock register is

$$H_{emit} = \sum_{d=0}^7 \left(J_d \otimes C_d + J_d^\dagger \otimes A_d \right), \quad A_d = C_d^\dagger.$$

For radiation energy equal to the matter gap, the total local energy commutator vanishes in the finite tests. The interaction is Hermitian, finite-frame covariant, and hard-core blocked if the destination rail is occupied. These results establish a lossless finite emission **architecture**. They do not select the scalar pulse angle, a source-supply law, or a unique many-body radiation correlator.

APPENDIX H. Final logical hierarchy of the integrated construction

Fields: Level | Exact / finite requirement | Emergent / matched / open requirement

- **Level:** Microscopic algebra — **Exact / finite requirement:** Finite matrices, exact unitary gates, finite Weyl composition — **Emergent / matched / open requirement:** Canonical variables only in low-phase charts
- **Level:** Causality — **Exact / finite requirement:** One BCC neighborhood per microscopic streaming/update phase — **Emergent / matched / open requirement:** Common relativistic light cone in the IR
- **Level:** Gauge/frame — **Exact / finite requirement:** Finite frame covariance and projectors in tested representations — **Emergent / matched / open requirement:** Unique final frame/address representation remains open
- **Level:** Matter — **Exact / finite requirement:** Exact flat BCC benchmark; constructive frame-covariant 8×4 carrier — **Emergent / matched / open requirement:** Universal interacting species/statistics sector remains open
- **Level:** Gravity — **Exact / finite requirement:** Bounded geometry registers, frame curvature, DeWitt structure, TT gates — **Emergent / matched / open requirement:** ADM/Einstein theory in the IR; G remains matched
- **Level:** HDA — **Exact / finite requirement:** Exact finite Weyl/deformation maps — **Emergent / matched / open requirement:** Strong code/Ward ADM closure; no exact whole-Hilbert infinitesimal algebra claimed
- **Level:** Strong gravity — **Exact / finite requirement:** Finite capacity and bounded local observables — **Emergent / matched / open requirement:** Effective regular profile and spacelike post-cap transition; unique future-core dynamics open
- **Level:** Rotation — **Exact / finite requirement:** Finite angular-momentum/frame-dragging constructions — **Emergent / matched / open requirement:** Kerr-like IR selected; microscopic Kerr uniqueness open
- **Level:** Radiation — **Exact / finite requirement:** Exact BCC streaming carrier and energy-conserving local source vertex — **Emergent / matched / open requirement:** Scalar pulse matched; multi-energy/species/greybody/source-supply dynamics open
- **Level:** Predictions — **Exact / finite requirement:** Structural/convergence tests are falsifiable — **Emergent / matched / open requirement:** No unique accessible quantitative astrophysical discriminator currently derived
- **Level:** Fundamental law — **Exact / finite requirement:** Finite all-register existence prototypes — **Emergent / matched / open requirement:** Unique U_{GCC}^{full} remains open

APPENDIX I. Selected E10-E31 numerical audit values

- E10: general-triad/finite-frame integrated curvature error fits approximately a^2 ; one representative integrated fit is $a^{2.03}$. A fixed 2T/T weak-link alphabet fails, while the 2T-gauge x near-identity transport representation succeeds under RG-refined transport resolution.
- E21: the actual BCC Floquet eigenphases agree with the analytic graph-mode law to numerical precision on finite periodic tests; fixed-CFL continuum dispersion approaches $\omega=c|k|$ with approximately $O(a^2)$ directional error.
- E27: on the isotropic Schwarzschild slice the reconstructed odd coefficient reaches $k_{eff}=3.99963$ at $a=0.05$, with $|k_{eff}-4|$ fitting approximately a^2 .
- E28: the direct TT curvature-Hessian eigenvalue ratio reaches 3.00122 at $a=0.05$; full target error scales approximately a^2 and polarization splitting is at roundoff.
- E29: the 18-phase integrated basis-control sector has maximum norm residual 2.8×10^{-15} , full inverse residual 5.0×10^{-15} , and zero work/cache leakage; coherent four-control-sector inversion is 4.4×10^{-15} .

- E30: common-metric matter+TT H-H closure scales as $dx^{1.99}$; a 10 percent TT gradient mismatch leaves an 8.5 percent floor. Geometry-aware H-D and D-D converge approximately as dx^2 .
- E31: physical H-H circuit error falls to 5.97×10^{-3} at $N=256$; H-D and D-D circuit fits are approximately $dx^{2.00}$ and $dx^{1.95}$. Full-state H-H remains about 78.8 percent because the target deformation is dominated by the metric motion absent from the current H circuit.

APPENDIX J. Schwarzschild spin-2 local chain

On the time-symmetric Schwarzschild spatial slice, the scalar curvature vanishes while the directional Ricci curvature does not. In an orthonormal frame, $E=R_{\hat{r}\hat{r}}=-2M/r^3$ and the tangential eigenvalues are $-E/2$. For a transverse-traceless tensor on the angular sheet, the local three-dimensional curvature endomorphism gives $K_{\text{curv}}(W)=3E W$. The local odd-parity master coefficient is therefore fixed by tensor geometry rather than by a fitted wave potential.

$$U_{RW} = f\left[\frac{l(l+1)}{r^2} + 3E\right] = f\left[\frac{l(l+1)}{r^2} - \frac{6M}{r^3}\right].$$

For $l=2$, taking this odd sector as independently fixed and imposing the nondegenerate Darboux relation on the minimal local even constraint class selects $C_2(y)=14y(3y+8)/(3y+4)^2$ and reproduces the standard Zerilli potential. The cited Schwarzschild perturbation literature supplies the continuum comparison only; the BCC curvature reconstruction and finite gate tests are the GCC-specific numerical steps [19-22].

APPENDIX K. E29 integrated scheduler

- 0: geometry-controlled matter update.
- 1: reversible geometry-difference work compute.
- 2: load/frame/transport reciprocal source update.
- 3: work uncompute.
- 4: capacity-conditioned geometry refresh.
- 5: oriented Link-loop curvature cache compute.
- 6: TT half-kinetic phase.
- 7-14: eight BCC perfect-matching TT gradient phases.
- 15: curvature-controlled spin-2 TT phase.
- 16: TT half-kinetic phase.
- 17: curvature-cache uncompute.

Every listed phase is finite and invertible. Strict microscopic locality is implemented by the matching/scheduler decomposition; the entire loop curvature need not be readable in one graph tick.

APPENDIX L. Smeared-deformation stopping point

A finite family of deformation circuits is stronger than a single Floquet unitary because the smearing data are explicit inputs. The the E10-E31 stage tests use small group commutators to extract the local algebra. The verified physical-sector relations are consistent with

$$\begin{aligned} [H[N], H[M]] &\rightarrow D\left[h^{ij}(N\partial_j M - M\partial_j N)\right], \\ [H[N], D[\xi]] &\rightarrow H[-L_\xi N], \quad [D[\xi], D[\eta]] \rightarrow D[[\xi, \eta]]. \end{aligned}$$

up to the convention-dependent sign of the generators and the stated refinement errors. At E31 the full-state H-H circuit still failed because the normal-deformation generator did not yet carry the complete geometry-momentum action. E31d supplied the finite DeWitt kinetic shear; E32-E44 then enlarged the nonlinear geometry sector, and E45 supplied the present strong infrared/code Ward closure. The remaining limitation is therefore not the old E31 stopping point but the absence of an exact whole-Hilbert infinitesimal HDA and of a uniquely selected final U_{GCC}^{full} .

APPENDIX M. Selected nonlinear HDA audit values (E32-E45)

The following values summarize representative checks; they are not fit parameters.

- **E39 classical spatial refinement:** representative H-H, H-D, and D-D errors over $N = 16$ -128 scale approximately as $a^{1.92}$, $a^{1.98}$, and $a^{1.97}$.
- **E39 quantum $\rightarrow$ same-discrete target, $N = 32, d = 5$:** H-H amplitude $\alpha \simeq 1.014$, shape error $\simeq 0.052$, cosine $\simeq 0.9987$; D-D shape error $\simeq 0.023$.
- **E40 joint limit, $N = 128, d = 5$:** H-H shape error $\simeq 1.36 \times 10^{-3}$; H-D $\simeq 5.68 \times 10^{-2}$; D-D $\simeq 1.92 \times 10^{-3}$.
- **E43 local full-six H-H register refinement:** shape error decreases from $\simeq 0.283$ at $d = 3$ to $\simeq 1.72 \times 10^{-2}$ at $d = 5$ and $\simeq 7.29 \times 10^{-4}$ at $d = 7$.
- **E44 classical generic-3D H-D/D-D refinement:** fitted spatial exponents are approximately 1.91 and 1.96.
- **E45 Ward H-D:** representative shape errors are $\simeq 2.73 \times 10^{-2}$ at $d = 5$, 1.96×10^{-2} at $d = 7$, 1.89×10^{-3} at $d = 9$, and 1.54×10^{-6} at $d = 15$.
- **E45 random-smearing robustness, $d = 9$:** H-D shape error $\simeq 3.05 \times 10^{-5}$ and D-D $\simeq 1.51 \times 10^{-4}$ in the tested coarse spatial ensemble.

These results support a controlled IR/code closure, not an exact microscopic continuum constraint algebra.

APPENDIX N. Selected strong-gravity and rotation values (E46-E49)

For the Planck-density effective core,

$$L_{core} = \sqrt{\frac{3c^2}{8\pi G\rho_p}} \simeq 0.345494 \ell_p.$$

The simple homogeneous horizon/cap ordering gives a crossover mass $M_{crit} \simeq 0.173 m_p$. Astrophysical masses satisfy $R_H/R_{cap} \gg 1$, so the cap lies deep inside the ordinary outer horizon.

For the E46 regular mass profile, the horizon-pair threshold is $\mu_{crit} = 3/2^{2/3} \simeq 1.88988$. In the astrophysical large-mass branch the effective spherical inner scale approaches $0.345494 \ell_p$. This sub-grid result is **not** the Kerr inner-horizon scale.

The E47 inner-horizon audit gives an astrophysical limiting surface-gravity scale $\kappa_t t_p \simeq 2.8944$, corresponding to an e-fold time $\simeq 0.3455 t_p$ in the effective spherical branch. Power-law tails therefore enter a nonlinear/capacity regime after only tens to hundreds of Planck times for extremely small seeds.

In Kerr,

$$r_{\pm} = r_g \left(1 \pm \sqrt{1 - \chi^2} \right),$$

so the inner horizon is macroscopic for ordinary spin. The E49 result is instead that the final mass-inflation-to-capacity layer is sub-grid. A Planck-core localization of all astrophysical angular momentum is impossible by the diagnostic $J/(McL_{core}) \gg 1$, requiring extended frame/Link support.

APPENDIX O. Selected radiation and evaporation values (E50-E55)

The final frame-covariant radiation carrier uses eight hard-core rails, giving local radiation dimension $2^8 = 256$ and combined local matter-radiation dimension $32 \times 256 = 8192$. Exact matrix-free tests give Hermiticity, energy-commutator, and finite-frame residuals at the 10^{-15} level.

Spectral continuity fixes

$$\kappa = 0.622494979899437, \quad \Delta = 1.244989959798873.$$

At $\rho = L_{core}$,

$$\frac{E_{\infty}}{k_B T_H} = 2.702628890469.$$

If the scalar pulse is matched to the standard Schwarzschild luminosity coefficient,

$$p = 2.231164802507 \times 10^{-3}, \quad \theta = 4.725278981575 \times 10^{-2} \text{ rad.}$$

The corresponding one-source mean first-emission waiting time is

$$\langle T \rangle = \frac{1}{p} \simeq 448.20$$

local attempts, with median 311. The area/entropy bookkeeping ratio is

$$\frac{N_{\text{emit}}}{A/\ell_p^2} = \frac{1}{4(2.702628890469)} \simeq 0.09250,$$

or one characteristic one-shot emission event per $\simeq 10.81$ Planck-area cells. These numbers belong to the **IR-matched single-gap model**, not to an independently normalized full Hawking spectrum.

APPENDIX P. Final claim ledger: statements that must remain limited

The following statements are explicitly **not** established by this paper:

1. A unique fundamental U_{GCC}^{full} .
2. An exact infinitesimal whole-Hilbert HDA on finite registers.
3. An independent microscopic derivation of Newton's constant.
4. A density-cap theorem forcing a bounce.
5. A dynamically stable classical inner Cauchy horizon.
6. A unique post-cap future-core geometry.
7. A microscopic derivation of the exact Kerr quadrupole and extremality relation.
8. Robust observable GW echoes from the deep core.
9. An independently predicted Hawking luminosity coefficient.
10. A unique thermal/greybody Hawking spectrum or a unique Page-time radiation correlator.

The strongest safe summary is: **GCC is a finite-QCA quantum-gravity candidate framework with explicit constructive realizations and substantial controlled infrared support, together with sharply identified microscopic selection and phenomenology gaps.**

Data and code availability

The E-series verification scripts and JSON summaries accompanying this manuscript are part of the reproducibility record. They are organized by stage number (E10-E56) and separate constructive tests from negative controls. The mathematical claims in the paper are intentionally stated at the level supported by those tests; the scripts do not substitute for analytic derivation, but they provide direct finite-dimensional and refinement checks of the stated constructions.

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